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Responding to the China challenge: Diversification and de-risking in new energy supply chains

Introduction 2
Michal Meidan, Philip Andrews-Speed, Anders Hove

China's grip on critical mineral and material supply: is it being weakened? 5
Philip Andrews-Speed

China critical minerals dominance: can innovation help? 9
Anders Hove

We need to talk about graphite: the geopolitics and economics of the battery anode value chain 12
Ahmed Mehdi

Indonesia's energy transition paradox: powering the EV battery supply chain with coal 18
Diwangkara Bagus Nugraha, Stein Kristiansen, Indra Overland

Securing the future: Japan's quest for critical minerals in the shadow of China 21
Parul Bakshi

US–African critical minerals collaboration: reducing trade frictions while enhancing value addition 25
Brad Simmons

The geopolitics and challenges of mining in Australia 28
Ian Satchwell and John Coyne

Latin America's positioning in critical mineral geopolitics 32
Tom Moerenhout and Victoria Barreto Vieira do Prado

Assessing the vulnerabilities and risks of China's role in wind power: A report from Sweden 36
Henrik Wachtmeister

Is China's solar PV manufacturing dominance a risk to the UK? 41
Dan Marks

CONTRIBUTORS TO THIS ISSUE 47

INTRODUCTION

Michal Meidan, Philip Andrews-Speed, Anders Hove

As tensions between the West and China deepen, critical minerals and materials for strategic technologies have become one of the battlegrounds. The technologies include military, information, and low-carbon energy applications. China's centrality to the supply of some of these critical minerals and materials has prompted Western governments to put in place countermeasures to ease their dependence on China. But as this issue of the *Oxford Energy Forum* explores, reducing China's role is more easily said than done.

The authors in this issue unpack China's dominance throughout the value chain, looking at some of the policies and corporate strategies that have enabled this. While much of the focus is on government responses to China, especially in minerals and materials, the analyses in this issue touch on the entire supply chain. While they do not cover all aspects of the value chains for solar cells, wind turbines, and lithium-ion batteries, nor all the commodities and the geographies where they are mined; but the complexity and linkages within these supply chains warrant a broader look, beyond the minerals alone. Miners have grappled traditionally with long lead times from mine to market, uneven regulatory frameworks, and environmental, social and governance (ESG) concerns, all of which still plague investments. But financing and investment decisions are complicated by de-risking policies that seek to encourage non-Chinese supplies. This is because the vast majority of minerals, metals, and ores are sent to China for refining and processing. Moreover, China's position as a monopoly supplier for some minerals and a near-monopsony buyer for several others gives it extraordinary market power. It has used this power to influence prices, which in turn has complicated investment decisions in both mining and refining. Finally, investment decisions around new mines also rely on the outlook for demand, which is in turn informed by technological innovation and advances in chemistries. But here too, China is also a front runner.

The China playbook

The *Forum* begins with an article by *Philip Andrews-Speed***,** arguing that diversifying supplies away from China will be extremely challenging in the coming decade, for the reasons discussed above: the challenges inherent in mining and China's pricing power. Investments in midstream facilities face similar challenges, although lead times tend to be shorter. Emissions and waste from these plants are often toxic, and their consumption of energy and water is usually large. In light of this, Andrews-Speed argues that innovation could be a more effective way to reduce or obviate the need for critical minerals in key technologies.

Anders Hove also suggests that a focus on supplies overshadows innovation as a solution to reducing dependency on China. But, as Hove notes, China is also a leading player on innovation, with efforts in the country running the gamut from process-oriented improvements to lab-stage technologies, from end-user design to upstream minerals processing.

Both *Hove and Andrews-Speed* stress supply chain integration. From a policy perspective, the Chinese government has supported firms' efforts to invest in minerals and ores overseas. Hove discusses Chinese firms' hybrid approach of vertical integration and intense innovation. These present a contrast with the strategies of many firms in advanced economies, which have tended to focus on disaggregating production to reduce costs. While Western mining companies focus on their most profitable, high-volume mining activities, carmakers and battery manufacturers seek to obtain inputs from the cheapest source globally. As a result, miners struggle to compete with Chinese investments in relatively low-volume critical minerals. Downstream players find themselves competing with Chinese firms for the offtake of critical minerals or investing in risky technology that may or may not pay off. An additional observation is that innovation goes hand-in-hand with deployment; and whereas Chinese players are confident domestic demand will continue to boom, Western players are waiting for the customer to catch up or supplier input prices to fall. This not only slows investment in the upstream but hinders innovation across the supply chain, putting Chinese players further in the lead.

These practices are captured neatly in *Ahmed Mehdi's* analysis of graphite, the largest component by weight in a battery pack for anodes and a key feedstock in the production of anode active material. Graphite is a useful case study because of China's vice-like grip on graphite markets and the evolving corporate playbook: Chinese firms have used their market dominance to squeeze out rivals, both domestic and international. A combination of vertical integration and feedstock switching in 2021–2022 helped consolidate Chinese anode players' global position. This was further aided by the Chinese government's tightening of export controls in late 2023.

Chinese companies are accelerating their push to exploit overseas mineral deposits, and Indonesia is one example. Nickel is a key ingredient of nickel-manganese-cobalt oxide battery cathodes, and the production of these cathodes is highly energy intensive. Indonesia has become the world's dominant supplier of nickel ore (52 per cent in 2023) and refined nickel (37 per cent

in 2023) over the last few years. This resulted from government policies to promote the extraction of nickel ore and to restrict the export of unprocessed ore in order to boost the nation's added value from this supply chain. The country now hosts 54 nickel smelters, five times that in 2020. Of these, seven produce materials for lithium-ion batteries, namely nickel matte and mixed hydroxide precipitate. As *Diwangkara Bagus Nugraha, Stein Kristiansen, and Indra Overland* document, Chinese companies have a significant shareholding in all but one of these seven smelters.

Whilst these policies have benefitted Indonesia, they have come at three costs. First, the deep involvement of Chinese companies in both mining and smelting further extends China's control of battery supply chains. Second, the use of coal-fired generators to power the smelters adds to Indonesia's already rising level of carbon emissions. Finally, the combination of these two factors results in very low production costs, causing many nickel operations in other countries to struggle.

Although the construction of grid-connected coal-fired power plants has been banned, nickel smelters are allowed to build captive coal-fired plants because these smelters are considered to be 'national strategic projects'. No steps have been taken yet to decarbonise these captive plants, and only 12 of the 54 smelters disclose their carbon emissions. The European Union's recent battery regulation establishes a carbon footprint threshold by 2028. This should affect Indonesia's nickel industry. However, most of Indonesia's nickel exports go to China, and China does not regulate the emissions of overseas investments.

Consumer governments' efforts to de-risk

China's increased use of export controls (on gallium, germanium, antimony, and graphite) as well as growing financial pressure on Western clean-tech firms and concerns about China's ability to use its dominance to pressure Western governments have led to a myriad of government responses. The US government has been amongst the most active through the 2022 Inflation Reduction Act, which supports investment and guides procurement along the full length of the supply chain, including mining and mineral processing. Resource-rich Australia and Canada are supporting investment in mining and processing for export. The European Union's Critical Raw Materials Act of 2024 includes 2030 targets for mineral extraction, processing, and recycling, as well as limiting over-dependence on a single third country. Realising that international cooperation was needed, the United States led the establishment of the Minerals Security Partnership (MSP) in June 2022 to 'bolster critical mineral supply chains essential for the clean energy transition'. The meeting in September 2024 saw the establishment of the MSP Finance Network, which includes development finance institutions and export credit agencies to support investment in critical mineral projects across the world. But are these efforts effective?

Japan's supply chain vulnerabilities came to the forefront during the 2010 diplomatic tensions with China, following a maritime incident near the Diaoyu/Senkaku Islands. *Parul Bakshi* reviews the incident that became the poster child of China's economic coercion, given that Beijing imposed an informal ban on the export of rare earth minerals to Japan. The embargo caused significant disruptions especially in the Japanese automotive sector, which relied heavily on these materials for essential components like magnets. While trade normalized after the resolution of the incident, the prices of rare earths surged tenfold within a year, highlighting the precarious nature of Japan's supply chain dependency.

The Japanese government developed a comprehensive strategy aimed at mitigating future risks associated with single-source dependency. This included development of alternative technologies that reduce the use of rare earths; promotion of recycling initiatives to recover rare earths from used products; strengthening capacities to invest in and develop rare earth mines globally, particularly in Australia, which emerged as a key partner; and establishing stockpiles of critical minerals to buffer against supply disruptions. Japan has effectively reduced its reliance on Chinese rare earth imports from 90 per cent to 60 per cent since the incident, and halved domestic consumption of these materials by 2023 compared to 2010.

In 2023, Japan and the United States signed an agreement focused on securing and diversifying supply chains for minerals for clean vehicle batteries. The Japanese government has also initiated collaboration with a number of countries in Asia and Europe, investing in mineral supply chains, research centres as well as joint investments in Africa. *Bakshi* argues that despite the risk that Japan's friend-shoring efforts could exacerbate economic disparities and inflate prices, its policy efforts have sought to balance economic, environmental, and security goals. But Bakshi also stresses that Japan must recognise the importance of integrating labour rights and environmental considerations into its development strategies, ensuring sustainable and equitable practices.

The US Inflation Reduction Act has been the largest national initiative to combat China's dominance of clean energy supply chains. A key policy instrument is the tax credit. However, the scope of this credit was expanded from manufacturing to include domestic ore extraction only in October 2024. Whilst this marks a major step forward, *Brad Simmons* explains that US ore reserves will be insufficient to match demand in the short term. Therefore, it will be necessary for the government to take additional steps to secure mineral supplies from other parts of the world such as Africa.

The Lobito Corridor project, backed by the G7's Partnership for Global Infrastructure Investment, is intended to transport minerals from Zambia and the Democratic Republic of the Congo to the Atlantic coast. Simmons proposes two additional initiatives directed at Africa. The first is to work with Morocco, which hosts vast reserves of phosphate that can be used to manufacture lithium iron phosphate batteries. Not only would this approach meet the requirements posed by the Inflation Reduction Act for sourcing nickel, but Morocco is the only African nation with a free trade agreement with the United States. The second proposal is to build a strategic partnership between the United States, the European Union, and the African Union focused on innovation in battery materials and components.

Producer countries' efforts to diversify

The view from producer countries is equally complex. On one hand, they are looking to capitalise on their resources as competition between China and the West deepens. China is for many a large offtaker and investor in mining. Meanwhile, the US is a key commercial and strategic partner, and the Inflation Reduction Act offers new opportunities. But Western policy and corporate overtures are sometimes caught between different policy priorities and struggle to compete with the Chinese playbook.

As a mineral-rich country, Australia has the opportunity to take advantage of the growing demand for critical mineral ores and metals by developing new mines and refineries. In addition, its mining companies are active overseas. Together, these provide the opportunity for mineral and metal importers to reduce their dependence on Chinese suppliers. However, as *Ian Satchwell and John Coyne* explain, Australian companies are finding it tough to compete with their Chinese counterparts. The country hosts several rare earth deposits, only one of which has been developed and is in operation, at Mount Weld. Development of the others is being constrained by China's ability to keep rare earth prices low. A proposed rare earth refinery faces similar challenges. Likewise, Chinese nickel projects in Indonesia have driven down prices for this metal, forcing Australian projects to suspend operations. In Africa, two Australian-operated lithium mines have been expropriated and passed to Chinese companies.

However, China is not just a competitor to Australian mining interests, it is also a collaborator as a customer for Australian ores and an investor in Australian mines and processing plants. Australia therefore has to take care to balance a variety of interests. Policies have been introduced to promote domestic investment in mining and processing, and 27 bilateral and multilateral agreements have been signed with offtakers for Australian minerals and metals. Despite growing agreement between key actors on the need for cooperative action to address China's dominance, policy inconsistencies between them undermine effective implementation. For instance, the US financially supports Australian companies in developing critical minerals supply chains outside of China, with the aim of further processing in the US. In the meantime, the Australian government currently restricts its financial support to critical minerals mining and processing operations within Australia. A further challenge is to define and then ensure high standards of sustainability.

Latin America has long been known as an important source of mineral supplies. *Tom Moerenhout and Victoria Barreto Vieira do Prado* argue that the region has new opportunities and challenges as it looks to capitalise on its mineral wealth. Fundamentally, though, Latin America must find partners, not just offtakers.

Among critical minerals, the continent only achieves its potential for copper, tin, and zinc but it has substantial untapped reserves for nickel, graphite, lithium, and rare earths. While the African continent has proved historically more attractive for investors, the generally low standards of environmental and social governance are likely to dissuade many Western investors. In contrast, many Latin American countries are actively raising these standards. In addition, low-carbon energy is more plentiful than in Africa. Set against these advantages, the regulatory and fiscal regimes in Latin American countries have tended to be unattractive to foreign investors.

China has been the exception. It is not only the major destination for Latin America's mineral exports but also the single largest investor in the continent's mineral resources. Chinese companies are also moving downstream into battery and electric vehicle manufacturing, building local supply chains and creating value added in the region. The United States and the European Union have started to build partnerships with Latin American countries directed at cooperation in the sustainable exploitation of critical minerals and in an effort to gain influence in the region. But mining companies from the United States, Europe, and Australia have been slow to move in. Not only are investments in mining challenging, as discussed above, but these countries would need to invest in processing capacity in order to fully benefit from Latin American minerals. Further, the multilateral Minerals Security Partnership includes no countries from Latin America.

Still, great power competition ultimately serves Latin America as geopolitical stress could become an opportunity to raise investment for their respective clean energy industries. This is true as long as they can remain nonaligned and not get dragged too much into the deteriorating relations between the US and China.

Assessing the risks

But for consumers in the OECD countries, geopolitical stress and China's dominance of clean energy supply chains raises numerous security concerns. *Henrik Wachtmeister* presents a summary of a systematic assessment of six of these risks relating to wind power in Sweden. The analytical framework distinguishes threat, vulnerability, and risk. The risks Wachtmeister identifies relating to Chinese ownership are electricity supply cuts and market manipulation, information transfer, and indirect influence. Those arising from dependence on Chinese supply chains are export restrictions, IT sabotage, and again indirect influence.

Wachtmeister then applies this framework to assess the risk of electricity supply cuts in Sweden arising from Chinese ownership of wind farms and the risks of export restrictions. His analysis reveals that the current risk of power cuts is low, even though Chinese state-owned companies own 10.4 percent of Sweden's installed wind power capacity. This is because these plants contribute only 3.4 percent of national electricity supply. Also, the companies are tied to Power Purchase Agreements under which they would suffer penalties if they cut off supplies. However, the risks to supply would be greater if new investments by Chinese companies lack these agreements. The highest risk therefore relates to the wind supply chain rather than to wind farm ownership because of potential export restrictions that stem from China's dominance of supply chains for components and for rare earths. This is due to the high consequence and medium-to-high probability of such restrictions being imposed. This, in turn, will depend on the state of political and economic relations between the EU and China. Managing this and other risks will require careful balancing of measures to support domestic industries whilst still taking advantage of Chinese prowess.

In a similar vein*, Dan Marks* assesses the risks for the UK that stem from China's dominance in solar PV. China accounted for 71 per cent of global solar cell exports in 2023. Although China supplied more than 90 per cent of the UK's imports of solar PV equipment in 2023, the absolute value of these imports was much lower than the peak in 2014 as deployment of solar equipment in the UK has slowed. Although solar PV may supply no more than 15 percent of the nation's electricity at any point between now and 2050, significant capacity additions will still be required.

Any constraint on or interruption to the supply of solar PV equipment from China, whether or not directed specifically at the UK, would raise the cost of electricity to consumers but should have no significant effect on security of supply. In contrast, cyber attacks on inverters and control devices pose a direct threat to supply security. Whilst supply interruptions pose a risk for the UK's solar industry, that industry only employs about 7,000 people, though this could rise to 60,000 by 2035 if there is a boom in deployment.

Beyond the specific risks facing the UK, other concerns relate to the environmental and social governance of China's solar PV supply chains and the potential for China to use its dominance of this and other clean energy supply chains for geostrategic aims.

CHINA'S GRIP ON CRITICAL MINERAL AND MATERIAL SUPPLY: IS IT BEING WEAKENED?

Philip Andrews-Speed

As tensions between the West and China have risen, critical minerals and materials for strategic technologies have become one of the battlegrounds for the two parties. The technologies include military, information, and low-carbon-energy applications. China's dominance of the supply of some of these critical minerals and materials has persuaded Western governments to put in place countermeasures to ease their dependence on China. This paper assesses the likely effectiveness of these measures.

The criticality of a specific material or technology is generally assessed on the basis of the risk of interruption to supply and on the economic or security importance to an importing nation of such a disruption. In this context, market concentration is a key component of the risk of supply disruption, along with perceptions of the dominant market actors, the projected balance between supply and demand for the material or product, and the degree of transparency of the respective market.

In the context of the low-carbon energy transition, risks appear along the full supply chain from mineral ores and refined metals to intermediate products and components to final products. Criticality is most often applied to mineral ores and their refined products. Most mineral ores have a relatively widespread geographic occurrence, but the economic viability of deposits is highly variable. For some minerals, only a limited number of mines may be operating at a given time and these may be geographically concentrated. Whilst the locations of ore deposits are defined by geology and immovable, ore processing plants to produce refined metals can be set up wherever economic, logistical, and political factors are favourable. Economies of scale are particularly important. As a result, the sources of supply of a refined metal are commonly much more concentrated than those of its primary ore. Today, China holds a strong and sometimes dominant position in the supply of a few mineral ores and several refined metals that are key inputs to clean energy appliances.

The governments of many countries have drawn up lists of minerals or materials that they consider to be critical or strategic. These lists vary between countries depending on their domestic natural resources and the demand of their industries. Box 1 lists metals relevant to the low-carbon energy transition that appear on several of these lists.

Box 1: Materials that appear on the critical mineral lists of many countries

Aluminium	Graphite	Platinum group metals
Antimony	Indium	Rare earth elements
Boron	Lithium	Silicon
Cadmium	Magnesium	Strontium
Chromium	Manganese	Tellurium
Cobalt	Molybdenum	Titanium
Copper	Nickel	Vanadium
Gallium	Niobium	Zinc
Germanium		

China's grip on critical mineral and material supplies

China's share of global production of mineral ores is small or modest in most cases (Table 1). The most notable exceptions are natural graphite, rare earths, and silicon, where the share is around 70 per cent or more. The situation for refined and processed metals is quite different, for China imports large quantities of ore from around the world to refine or process within its borders. As a result, it accounts for around 90 per cent or more of global production of gallium (a byproduct of bauxite processing), spherical graphite, manganese sulphate, magnetic rare earths, and polysilicon. The country also produces around 60 per cent or more the world's refined aluminium, cobalt, germanium, and indium (both byproducts of zinc ore processing), lithium carbonate, and tellurium (a byproduct of copper ore processing).

Table 1: China's share of global ore and metal production for selected critical materials (mainly for 2023, but some data for 2022)

Ore	China share	Refined/processed metal	China share
Antimony	48%		
Bauxite	23%	Aluminium	59%
		Cadmium	39%
Cobalt	1%	Cobalt	74%
Copper	8%	Copper	44%
		Gallium	98%
		Germanium	~ 60%
Graphite (natural)	77%	Spherical graphite	99%
Graphite (synthetic)	52%		
		Indium	66%
Lithium	18%	Lithium carbonate	74%
Manganese	4%	Manganese sulphate	97%
Molybdenum	42%		
Nickel	3%	Nickel	18%
Rare earths	69%	Magnetic rare earths	88%
Silicon	73%	Polysilicon	>90%
		Tellurium	67%
Zinc	33%		

Source: Principal sources: International Renewable Energy Agency (2024), *Geopolitics of the Energy Transition*, Geopolitics of the energy transition: Energy security (irena.org); United States Geological Survey (2024), *Mineral Commodity Summaries 2024*, https://pubs.usgs.gov/periodicals/mcs2024/mcs2024.pdf; International Energy Agency (2022), S*olar PV Global Supply Chains*, https://www.iea.org/reports/solar-pv-global-supply-chains

China's precise share of global mineral and metal production can vary from year to year either as part of a long-term trend or due to changes within or outside China. For example, the share of aluminium smelter production increased steadily from 48 per cent in 2014 to 59 per cent in 2022. Conversely, the nation's share of production of rare earth ores declined from 95 per cent in 2011 to 81 per cent in 2016 as the government tightened controls over mines.

China's 2016 National Plan for Mineral Resources classifies the country's mineral resources as 'strategic', 'advantageous', 'protected', or 'strategic emerging industry' minerals. For the different categories, the plan identifies where China needs to encourage exploration of minerals in short supply, regulate the amount of minerals that China can leverage to pursue strategic objectives, cut production of minerals with excess capacity, and ensure supply of minerals in strategic emerging industries. Two categories are relevant to this paper:

- metallic minerals—iron, chromium, copper, aluminium, gold, nickel, tungsten, tin, molybdenum, antimony, cobalt, lithium, rare earths, and zirconium;
- nonmetallic minerals—phosphorus, potash, crystalline graphite, and fluorite.[1]

China's growing grip abroad

To compensate for the paucity of ore reserves at home, Chinese companies have moved overseas. They have had long-standing involvement in a small number of countries such Guinea for bauxite, the Democratic Republic of Congo for cobalt, and South Africa for platinum group metals. More recently, they have expanded their overseas activities to mine bauxite and nickel in Indonesia and lithium in a trio of Latin American countries as well as Zimbabwe, among other mining projects (Table 2). In addition, these and other Chinese companies are increasing their involvement in the processing of mineral ores, both in those countries where they are mining and those where they are not. Most notable is Chinese involvement in nickel processing in Indonesia, where the sudden surge of output combined with slowing demand for electric vehicles has undermined nickel prices and threatens the viability of multinational firms involved in nickel mining and refining.

Table 2: Examples of countries where China is deeply involved in the extraction and processing of critical minerals or has bilateral offtake agreements

Mineral	Country	Extraction	Processing	Offtake agreements
Aluminium	Indonesia	Yes	Planned	Yes
	Guinea	Yes		Yes
Cobalt	Democratic Republic of Congo	Yes		Yes
Copper	Chile		Planned	
	Democratic Republic of Congo	Yes		Yes
	Zambia	Yes	Yes	
Lithium	Chile	Yes	Planned	Yes
	Argentina	Yes	Planned	
	Bolivia	Yes		Yes
	Australia			Yes
	Zimbabwe	Yes	Yes	
Manganese	Gabon			Yes
Nickel	Indonesia	Yes	Yes	Yes
	Philippines		Planned	
Platinum group metals	South Africa	Yes		Yes

In this context, it is important to note that China's overseas activities have evolved and continue to change based on host country preferences and changes in domestic and international market dynamics. While much of China's outbound investments in minerals and mining in the early 2000s was supported by the Chinese government or state-owned policy banks, there has also been considerable activity driven by corporate interests and initiated by both state-owned and private companies.

[1] China Ministry of Natural Resources (2016, November), 'China's 24 strategic minerals', https://www.gov.cn/xinwen/2016-11/30/content_5140509.htm.

Sources of risk posed by China's dominant position

Disruptions to supplies of ores or materials for which China holds a strong position may be unintentional or intentional. Unintentional disruptions arising within China include natural events and disasters such as floods and droughts, as well as industrial accidents and epidemics. Any of these could significantly reduce the supply of ores and materials from domestic sources for a period. In such circumstances, the government is likely to prioritise supplies to domestic industries at the cost of exports. In an overseas example, China diverted graphite exports away from Sweden and towards Hungary to support a Chinese battery factory there.

The same is likely to apply to any country that has a significant hold on a critical mineral or material supply chain. If such a country supplies mineral ores to China for refining there, then a disruption of ore supply in that country will result in a disruption of processed materials supply in China. In addition to natural events, accidents, and epidemics, this type of disruption may be caused by regulatory action by the host government, independent from any actions by China's government.

Intentional disruptions are of two main types: export controls and economic statecraft. Over the last year or so, the government has deployed export controls on rare earths, graphite, gallium, germanium, and antimony. These controls do not automatically lead to a reduction of export volumes. Rather, they give the government levers to impose restrictions at short notice. In contrast, the government has banned the export of equipment for extracting and separating rare earths. Economic statecraft involves targeting a specific country. An early prominent example was the reduction of exports of rare earths to Japan in 2010 following a fishing dispute. While exports to Japan fell, flows to other countries, including Australia and the UK, declined too.

The countermeasures of the West

Japan was the only country that took some effective action after China reduced rare earth exports in 2010. However, since 2020, most high-income nations have started to take steps to reduce their dependence on Chinese supplies of critical minerals and materials. These actions have taken place at the national, regional, and global levels.

The US government has been amongst the most active through the 2022 Inflation Reduction Act, which supports investment and guides procurement along the full length of the supply chain, including mining and mineral processing.[2] Resource-rich Australia and Canada are supporting investment in mining and processing for export.[3] The EU's Critical Raw Materials Act of 2024 includes 2030 targets for mineral extraction, processing, and recycling, as well as limiting overdependence on a single third country.[4]

Realising that international cooperation was needed, the United States led the establishment of Minerals Security Partnership (MSP) in June 2022 to 'bolster critical mineral supply chains essential for the clean energy transition'. A meeting in September 2024 saw the establishment of the MSP Finance Network, which includes development finance institutions and export credit agencies to support investment in critical mineral projects across the world.

The likely effectiveness of these measures

Several factors will act to constrain the effectiveness of these measures, despite the financial and political capital being deployed in their support. The first set of constraints relates to the realities of the mineral industry. The development of a new mine anywhere faces several interrelated constraints. First is the need for investors who have access to capital and are prepared to risk fluctuations in commodity prices and host government regulations. Second is the time needed to move from discovery through feasibility study to first production. Even under favourable conditions, this can vary from five years or so for lithium mines to more than 10 years for copper or nickel mines. Third are the negative environmental impacts of mining. These include the use and degradation of land, pollution of air and water, damage to biodiversity, and emission of greenhouse gases. The final constraint relates to the effects on local communities, including their livelihoods and health. The consequences of these and other factors vary between countries and projects. But together they result in high costs, risks, and lead times for most projects. Similar challenges face mineral processing facilities. Though construction times will generally be less than for mines, emissions and waste from these plants are often toxic and their consumption of energy and water is usually large.

[2] Meaghan Connors, J. Paul Forrester, and Kevin L. Shaw (2022, 29 September), 'Strengthening the US supply chain for critical minerals and the Inflation Reduction Act—opportunities and challenges', *Mayer Brown*, https://www.mayerbrown.com/en/perspectives-events/publications/2022/09/strengthening-the-us-supply-chain-for-critical-minerals-and-the-inflation-reduction-act-opportunities-and-challenges.

[3] Australian Government, Department of Industry, Science and Resources (2023, 20 June), *Critical Minerals Strategy 2023–2030*, https://www.industry.gov.au/publications/critical-minerals-strategy-2023-2030; Government of Canada, Ministry of Natural Resources (2022), *The Canadian Critical Minerals Strategy*, https://www.canada.ca/en/campaign/critical-minerals-in-canada/canadian-critical-minerals-strategy.html.

[4] European Commission (2024, 11 April), *European Critical Raw Materials Act*, Regulation - EU - 2024/1252 - EN - EUR-Lex (europa.eu).

As a result, obtaining societal and governmental approval for mining and processing projects in most high-income and many middle-income countries is likely to be costly and time consuming and may ultimately be unsuccessful. The ongoing example in Europe is Rio Tinto's plan to open a lithium mine in Serbia, where strong public opposition is delaying the start of construction despite strong government support and the potential for the mine to transform the economy. A plan to develop a rare earth mine in Arctic Sweden is also facing strong objections. In the United States, tougher environmental regulations are delaying project approvals. The country reportedly has the second longest average time for a mine to progress from discovery to first production, at 29 years. Whilst approval periods may be shorter, the challenge in many mineral-rich countries is to ensure that environmental, social, and governance systems meet international standards. Indonesia is a case where rapid Chinese investment in nickel mining and processing has led to extensive environmental damage.

The second set of constraints originate in China. The country's strong position in the supply of a few critical mineral ores and several refined metals and materials gives it great influence over international prices. A small change in the share of its production that is exported can have a disproportionate impact on prices. The causes of such fluctuations in export volumes can be unintentional or intentional, as discussed above. The combination of these fluctuations with short-term variations in global demand results in highly unpredictable prices for mineral ores and metals. This is especially the case for critical minerals which are traded in small volumes on illiquid and non-transparent markets. The Mountain Pass rare earth mine in California is an example of a mine that has periodically closed due to such price fluctuations along with environmental regulation. Only with Department of Defense financial support was it able to build its own processing plant. Before that, the ores had been sent to China for processing.

For these reasons, the likelihood is small that these measures to promote mining and ore processing outside China will have a substantial impact on the country's currently strong or dominant positions in the supply chains for certain critical minerals and mineral products, at least over the next 10 years or even longer. However, they may make a modest contribution to local supplies. More effective may be innovation to reduce or obviate the need for critical minerals in key technologies. These include the development of sodium-ion batteries and permanent magnets free of rare earths. The problem here is that China itself is one of the leaders is developing and bringing to market these technologies. As a result, these innovations may just reinforce China's dominance of clean energy equipment supply chains, despite a diminution in its dominance of critical mineral supplies.

CHINA CRITICAL MINERALS DOMINANCE: CAN INNOVATION HELP?

Anders Hove

China dominates the market for many of the essential minerals and materials needed to power the global transition to clean energy. This notably includes mining and processing of rare earths used in permanent magnets for wind and electric vehicle (EV) motors, processing of lithium and other minerals used in batteries, and production of silicon for solar panels. As world leaders seek to reduce dependence on China for these critical minerals, analysts have focused narrowly on forecasting the likely supply and demand for these minerals, and on measures to enhance non-Chinese supply. However, as Philip Andrews-Speed argues in this issue, supply-focused measures adopted by Western countries to date are unlikely to have a large impact on Chinese dominance of critical minerals over the next 10 years.

While alternatives to critical minerals are part of the supply-demand analysis - for example, lithium-iron-phosphate (LFP) batteries or sodium-ion (Na-ion) batteries as alternatives with fewer mineral supply constraints - the broader question of innovation as a solution is often omitted.

Not only is China dominant in critical minerals for the energy transition, it has also become a leading innovation player in clean energy, largely as a result of its robust domestic demand fed by a dynamic and hypercompetitive manufacturing sector. Innovations that reduce critical minerals dependence have already arrived - in China. The question then becomes, how did China achieve dominance of both critical minerals supplies and clean energy innovation? And if innovation in alternatives to critical mineral inputs is likely to help reduce minerals-related vulnerabilities, will this just be replaced by more dependence on China for those very alternatives?

Achieving dominance: scale and vertical integration

There are three primary reasons why China dominates the critical minerals used in clean energy technologies today. First, China's rapid scale-up of manufacturing in these technologies, including for domestic Chinese markets, made upstream integration essential. Second, China's long-term, policy-led focus on mastering and localising strategic industries supported this effort, helping companies weather boom-and-bust cycles, easing financing for risky upstream investments, and facilitating technology transfer

in key minerals processing fields. Third, policymakers, policy banks such as the China Development Bank, and state-owned industry actively organised 'going-out' policies to facilitate large resource deals in the developing world - often in ways that couldn't be matched by other countries, not even by the largest private companies in the global mining and resource industries.

China has led the world in clean energy manufacturing for so long that it has come to seem almost inevitable - even though this was hardly foreseeable two decades ago. In 2023, the country sold 8.9 million passenger plug-in EVs, including plug-in hybrids, and in one month sales exceeded 1 million for the first time. In other words, plug-in vehicles in China in a single month exceeded the entire US EV market for the full year of 2023, and accounted for an astonishing two-thirds of the global market for such vehicles. As of mid-2024, EVs have surpassed a 50 per cent market share in new passenger vehicle sales on a monthly basis - not a small feat, given that China is the world's largest car market. Similarly, for solar energy, China not only dominates manufacturing of photovoltaic (PV) cells and modules, but its domestic installation of solar in 2023 amounted to around 200 gigawatts, more than the entire US solar PV installed nationwide and equivalent to half the world market for solar. Installations are on track to top this record figure in 2024.

In both EVs and solar, China not only leads by a wide margin, but has experienced exponential growth sustained over several years, fuelled by fierce domestic competition among manufacturers keen to achieve scale and market share. In this situation - rapidly growing domestic demand, high returns to scale, and low margins - upstream integration would be almost essential to individual businesses and the industry as a whole.

Geographical concentration of upstream suppliers within China was essential to the rapid expansion of China's wind and solar industries. Often, regional or local clusters played a role, such as the Pearl River Delta and Yangtze River Delta for solar PV and northern China for wind. Policy and technology characteristics each helped create this concentration: in wind, Chinese domestic content requirements for local projects encouraged foreign manufacturers to set up production in China and to transfer technology, whereas in solar PV, Chinese startups leveraged turnkey production lines imported from abroad.[5] Upstream integration has also been important to the development and scale-up of the battery and EV sectors, where Chinese companies that began as start-ups just over a decade ago now dominate the market for making batteries and pose an imminent threat to leading global carmakers. As just one example, BYD famously sources 90 per cent of parts for its EVs from in-house.[6] Other firms rely on a well-developed local supplier network, such as CATL, which has helped develop a battery industry cluster in and around Fujian province.

Several factors contributed to China's vertical integration in wind, solar, batteries, and EVs, and a full discussion of these is beyond the scope of this paper. As regards critical minerals, however, the important point is that vertical integration in China has not been limited to the immediate fields of manufacturing. Over the past decade, not only has China accelerated the approval of lithium mining operations and rapidly expanded lithium output and processing, but EV and battery companies have invested in upstream production. Over 400 Chinese companies are now involved in battery materials processing.[7] While the Chinese government has promoted the mining sector for years, a major motivation for companies entering the upstream was the need to keep ahead of the domestic competition and secure reliable access to raw material supplies.

Demand and domestic competition as catalysts for innovation

China's rapidly scaling manufacturing capacity for wind, solar, and batteries has led Chinese firms to compete both to secure upstream minerals supplies and to innovate to reduce costs. Innovation in China runs the gamut, from process-oriented improvements to lab-stage technologies and from end-user design to upstream minerals processing. All are relevant to critical minerals, because every such innovation has the potential to reduce global critical minerals demand.

[5] Jonas Nahm and Edward S. Steinfeld (2014), 'Scale-up nation: China's specialization in innovative manufacturing', *World Development* 54, https://doi.org/10.1016/j.worlddev.2013.09.003; Jeffrey Ball (2017), 'The new solar system', Steyer-Taylor Center for Energy Policy and Finance, Stanford Law School, https://law.stanford.edu/publications/the-new-solar-system/; Joanna Lewis (2013), 'The role of foreign technology in China's wind power industry development', in *Green Innovation in China: China's Wind Power Industry and the Global Transition to a Low-Carbon Economy*, Columbia University Press, https://doi.org/10.7312/lewi15330; Aitong Li et al. (2023), 'Gradual improvement and reactive intervention: China's policy pathway for developing the wind power industry', *Renewable Energy* 216, https://doi.org/10.1016/j.renene.2023.119068.

[6] Edward White and Peter Campbell (2022, 8 July), 'The "nutty" professor behind rise of China's electric vehicle giant', *Financial Times*, https://www.ft.com/content/367af909-5cdb-4780-a577-b84f815bcddd; Umesh Bhagchandani (2022, 8 August), 'Forget Elon Musk, meet Wang Chuanfu: the Chinese billionaire has dethroned Tesla with BYD's 2022 electric car sales and was backed by Warren Buffett—but comes from a family of farmers', *South China Morning Post*, https://www.scmp.com/magazines/style/celebrity/article/3188247/forget-elon-musk-meet-wang-chuanfu-chinese-billionaire.

[7] Huiwen Gong and Allan Dahl Andersen (2024), 'The role of material resources for rapid technology diffusion in net-zero transitions: insights from EV lithium-ion battery Technological Innovation System in China', *Technological Forecasting & Social Change* 200, https://doi.org/10.1016/j.techfore.2023.123141.

Unsurprisingly, the area where China's innovation has taken a clear lead is in manufacturing - in short, in process-oriented innovation. This type of innovation has helped Chinese firms compete in clean energy sectors where the initial technology was developed elsewhere. In solar and battery manufacturing, Chinese players entered the field after a standardised and modular product design had already become dominant; therefore, reducing costs and beating out competition was largely a matter of scaling up manufacturing and reducing inefficiency and materials losses during production. To take solar PV as an example, production know-how is critical to reducing losses of solar-grade silicon during each stage of the process, from sawing ingots into wafers, to making wafers thinner, to avoiding broken or faulty cells and modules.[8]

Beyond mastery of process-oriented know-how, China's scale and heavy investment in R&D means that Chinese players are increasingly able to lead in technologies that reduce materials consumption. This includes both technology-driven improvements in efficiency and investments in new materials that dramatically reduce or avoid critical materials altogether.

On efficiency, Chinese solar and battery manufacturers are now technology leaders. Chinese firms dominate the production of the most advanced solar technologies, with new record solar conversion efficiencies being announced regularly by Chinese manufacturers, including for materials still in the lab stage such as perovskite tandem cells,[9] as well as for fully commercialised technologies such as bifacial, PERC (passivated emitter and rear contact), and n-type monocrystalline PV.[10] In batteries, Chinese players are holding their own in the patent and R&D race versus Japanese, Korean, and US competitors. CATL and BYD, the leading battery players, regularly announce cutting-edge new battery technologies with market-beating efficiencies - for example, CATL's condensed battery that offers energy density of 500 Wh/kg, or BYD's Blade battery, which offers greater energy density and reduced cost due to its cell-to-pack design.[11] Again, steadily improved efficiency helps reduce materials consumption.

Perhaps more important still, Chinese players are leading the race to find alternative technologies that reduce or displace demand for critical minerals. This process is already in evidence with the case of LFP battery technology, which went from being an also-ran in the industry - surpassed in energy density by the more advanced nickel-manganese-cobalt battery chemistry - to taking the majority of battery sales in the huge Chinese EV market.[12] Chinese players didn't invent LFP: that milestone goes to the Nobel Prize-winning John Goodenough and researchers at University of Texas-Austin and Oxford University. Instead, Chinese companies licensed the technology and scaled it up, in some cases benefitting from state help, such as the invalidation of key patents or the acquisition of US battery startup A123 by Wanxiang.[13] Whereas Western carmakers and Asian battery makers focused on the high end, seeing EVs as a premium product, Chinese players saw an opportunity to compete with lower-priced offerings. Through ongoing innovation, Chinese manufacturers were able to bring LFP-equipped EVs to a level where the price-to-performance ratio was compelling in many mid-range offerings, including notably the made-in-China Tesla Model 3 as well as many US-made standard-range Tesla vehicles.[14] While competing technologies such as nickel-manganese-cobalt batteries have also managed to reduce consumption of critical minerals like cobalt, the impact of China's success with LFP on materials demand for batteries is undoubtedly far larger.

The same process is now being repeated with Na-ion batteries, another technology long discounted by leading players for its relatively low efficiency, despite its potential for reducing global demand for lithium. While Na-ion batteries have some potential for vehicles, their near-term market is in the field of stationary energy storage. Once again, Chinese players are taking the lead,

[8] Anu Bhambhani (2024, 15 March), 'Chinese solar industry to get bigger & better in 2024', *Taiyang News*, https://taiyangnews.info/markets/chinese-solar-industry-to-get-bigger-better-in-2024; LONGi (2024, 29 April), 'LONGi brings next generation TaiRay silicon wafer products to the global PV market', https://www.longi.com/en/news/silicon-new-product-release/.

[9] Jinko Solar (2023, 10 November), 'Three-peat victory! JinkoSolar sets new records for cell, module, and tandem efficiency successively', https://www.jinkosolar.com/en/site/newsdetail/2058.

[10] National Renewable Energy Laboratory (n.d., accessed 8 March 2024), 'Best research-cell efficiency chart', Photovoltaic Research, https://www.nrel.gov/pv/cell-efficiency.html.

[11] Lei Kang (2024, 25 June), 'CATL expects its batteries to power electric aircraft with up to 3,000 km range', *CNEV Post*, https://cnevpost.com/2024/06/25/catl-batteries-to-power-electric-aircraft-3000-km/; Peter Johnson (2024, 8 April), 'BYD is launching its next-gen Blade EV battery soon with more range and even lower cost', *Electrek*, https://electrek.co/2024/04/08/byd-launch-next-gen-ev-battery-more-range-lower-price/.

[12] International Energy Agency (2024), 'Global EV outlook: trends in electric vehicle batteries', https://www.iea.org/reports/global-ev-outlook-2024/trends-in-electric-vehicle-batteries.

[13] Yuki Kamikawa and Matthew Brummer (2024, 7 February), 'Cross-national and cross-sectoral dynamics of innovation policies: the case of lithium-ion battery technology for electric vehicles in the US and China', *Technological Forecasting & Social Change* 201, https://doi.org/10.1016/j.techfore.2023.123021; Tom Hals and Ben Klayman (2013, 30 January), 'Chinese firm wins A123 despite U.S. tech transfer fears', Reuters, https://www.reuters.com/article/business/chinese-firm-wins-a123-despite-us-tech-transfer-fears-idUSBRE90S0JO/.

[14] Fred Lambert (2022, 22 April), 'Tesla is already using cobalt-free LFP batteries in half of its new cars produced', *Electrek*, https://electrek.co/2022/04/22/tesla-using-cobalt-free-lfp-batteries-in-half-new-cars-produced/.

with CATL and BYD both actively developing and marketing Na-ion products[15] - geared first and foremost for the huge domestic market for grid-scale batteries to back up the country's surging wind and solar capacity, but also for EVs, where the lower-energy density Na-ion cells can be paired with lithium-ion cells to offer cost-effective performance.

Last but not least, Chinese innovation has also played a role in reducing upstream costs for mineral extraction and processing. In China's hyper-competitive domestic market for clean energy technology, companies seek not only to lock in supply of key inputs, but also to improve upstream efficiencies to beat out the competition. The result has been improvements in age-old but still essential technology processes for extracting and refining ores into pure minerals.[16] Technology transfer and state policy have also played a role - and continue to do so, as shown by the recent state-owned company acquisition of the one remaining foreign-owned rare earths processing facility for making ultra-pure dysprosium.[17] The mineral is essential for permanent magnets used in EV batteries and wind turbines, as well as in high-performance chip-making.

Implications and conclusions

China's dominance of the clean energy industry, and its hybrid approach combining vertical integration with intense innovation, present a contrast with the strategies of many firms in advanced economies, which have tended to focus on disaggregating production to reduce costs. Mining companies focus on their most profitable, high-volume mining activities, carmakers and battery manufacturers seek to source inputs from the cheapest source globally. For miners, surging Chinese investment in relatively low-volume critical minerals results in low prices and low or negative returns, making competition with Chinese players seem futile. For downstream players that need critical mineral inputs, the lack of alternatives puts them in the position of either getting at the back of the line for critical minerals behind competing Chinese original equipment manufacturers, or investing in risky technology bets (such as solid-state batteries at Stellantis and Toyota) that may or may not pay off.

While the Chinese corporate strategy of vertical integration has succeeded, its core driver is policy. Although subsidies and planning targets are often cited as the main policies at play, there is more to the story. China's development approach is based on designating strategic emerging industries for long-term support, ultimately leading private players and local officials to channel resources to such technologies. This results in intense domestic competition and efforts such as vertical integration aimed at driving down prices to beat out competitors. This not only causes manufacturing overcapacity, but also spurs rapid learning-by-doing, innovation, and price declines that ultimately help stimulate demand more than subsidies or five-year-plan targets ever could.

Innovation has already proven effective in reducing the world's demand for critical minerals like cobalt and nickel while scaling up clean energy technologies. The problem is, innovation goes hand-in-hand with deployment; and whereas Chinese players are confident domestic demand will continue to boom, Western players often take a losing approach, waiting for the customer to 'catch up' or for supplier input prices to fall. This not only slows investment in the upstream but hinders innovation across the supply chain, putting Chinese players further in the lead.

WE NEED TO TALK ABOUT GRAPHITE: THE GEOPOLITICS AND ECONOMICS OF THE BATTERY ANODE VALUE CHAIN

Ahmed Mehdi

Any discussion on the economics and geopolitics of the energy transition cannot ignore the role of graphite. Used as a feedstock to produce electrodes for the steel industry and anode active material (AAM) for the battery industry, graphite's strategic importance is only set to grow, reinforced by two structural trends: rising global lithium-ion battery demand and growing use of electric arc furnace (EAF) technology in the global steel industry. In the battery market, cathode active materials—such as lithium, nickel, and cobalt—have dominated supply-side concerns among governments and automotive original equipment manufacturers (OEMs).

[15] Phate Zhang (2023, 20 April), 'CATL, BYD's sodium-ion batteries both to be in mass production within this year, report says', *CNEV Post*, https://cnevpost.com/2023/04/20/catl-byd-sodium-ion-batteries-mass-production-this-year-report/; Carla Westerheide (2023, 21 April), 'CATL and BYD to use sodium-ion batteries in EVs this year', *Electrive*, https://www.electrive.com/2023/04/21/catl-and-byd-to-use-sodium-ion-batteries-in-evs-this-year/.

[16] Huiwen Gong and Allan Dahl Andersen (2024), 'The role of material resources for rapid technology diffusion in net-zero transitions: insights from EV lithium-ion battery Technological Innovation System in China', *Technological Forecasting & Social Change* 200, https://doi.org/10.1016/j.techfore.2023.123141.

[17] Keith Bradsher (2024, 26 October), 'China tightens its hold on minerals needed to make computer chips', *New York Times*, https://www.nytimes.com/2024/10/26/business/china-critical-minerals-semiconductors.html.

This is not surprising. While graphite is the largest component by weight in a battery pack for anodes, AAM only accounts for 7–10 per cent of the cell cost. Lithium, on the other hand, can make or break cell cost competitiveness. At the same time, anode technology developments have lagged the cathode, with graphite's incumbent status remaining largely unchanged. And for good reasons, too: graphite provides high thermal and electrical conductivity, reasonable energy density, and strong cycling ability, and is low cost - the latter an important consideration for such a margin-sensitive industry as the global EV market.

So, why is it time to talk about graphite? First, geopolitics: in no other part of the global battery supply chain does China have such a vice-like grip. The reality today is that without Chinese anode material, there is no Western EV industry. Second, market structure and pricing: developments in the Chinese battery and steel market over the past two years have rippled across ex-China markets, creating calls for industry protection via tariffs and subsidies. With the West seeking to play catch-up, what is the reality governing this market and where next for the ex-China market?

Graphite is here to stay

In 2023, global graphite demand reached around 3.6 million metric tonnes, with most of the demand split across two major industries - graphite for high-purity electrodes used in steel production, and graphite for AAM used in the battery industry. Graphite electrode demand is expected to continue growing over the next decade as EAF penetration rates rise, particularly in China (where current EAF penetration rates are around 10 per cent but are expected to reach 25 per cent by 2040). Despite some wobbles to EV sales this year (particularly in Europe), battery AAM demand from the EV sector will still see growth of 20–25 per cent this year. By 2026, it is expected that graphite demand from the battery sector will comfortably take over demand from the steel sector, reinforcing graphite's strategic importance to the global automotive and energy storage market.

Graphite's role as a primary feedstock for battery AAM comes from its chemical structure. The large interstices between the carbon atoms prevent structural change during charge and discharge cycles. As a result, graphite provides not only long cycle life to the battery but also reasonable energy density for modern battery cell chemistries.

Nevertheless, questions have been raised around the threats to long-term graphite demand, primarily from three major areas: (1) pure silicon anodes, which do not use graphite, (2) sodium-ion (Na-ion) batteries, which do not use graphite but rather hard carbons as the primary anode feedstock, and (3) solid-state batteries - seen as the holy grail for next-generation batteries - which would replace graphite-based anodes with lithium metal.

While all three technologies are set to play a role in the future evolution of the battery supply chain, graphite's incumbent position is likely to remain. This is for three main reasons:

- First, small amounts of silicon (7–8 per cent) are already being used in today's market to improve the energy density of a battery cell. Where graphite can store one lithium atom per six carbon atoms, silicon alloys can store up to four lithium atoms per silicon atom. The specific capacity of silicon - the amount of electric charge (measured in milliampere hours, mAh) that can be delivered per gram of material - is 3,590 mAh/g, ~10 times greater than that of graphite, reinforcing the energy density gains that can be made. The technical hurdle for silicon, however, remains the expansion of silicon during charge/discharge. While graphite anodes may expand by around 7 per cent during charge/discharge, silicon anodes can expand anywhere between 250 and 300 per cent during charge, leading to fractures and premature cell failure. In short, silicon doping remains the name of the game for now.
- Second, interest in Na-ion batteries has grown since the lithium price hike of 2022 - a technology actively promoted by Chinese cellmakers as a hedge to lithium price volatility. The strong cycle life and safety of Na-ion has made the technology a widely coveted one, particularly by storage developers. The economics of Na-ion are highly compelling when lithium prices are above US$40kg/t, but less so in a US$10–20kg/t environment. This reinforces the point that Na-ion will certainly play a future role in the battery market but will be sensitive to the lithium price. At the same time, while Na-ion anodes cannot use graphite (due to the larger ionic size of sodium ions, which do not fit between the graphite anode layers) the replacement material, hard carbons, remains a small market which requires scaling.
- Finally, the solid-state battery market - which would replace graphite anodes with lithium metal - remains in its infancy. Not only has the supply chain not yet fully scaled, but ongoing challenges with the use of a solid electrolyte remain, particularly dendrite formation.

The synthetic and natural split

The battery anode market has two main types of supply: natural graphite AAM and synthetic graphite AAM. The natural anode supply chain starts with mined material known as natural flake graphite, which comes in various flake sizes and purity levels. The flake graphite undergoes mechanical treatment such as spheronisation and purification to become natural AAM.

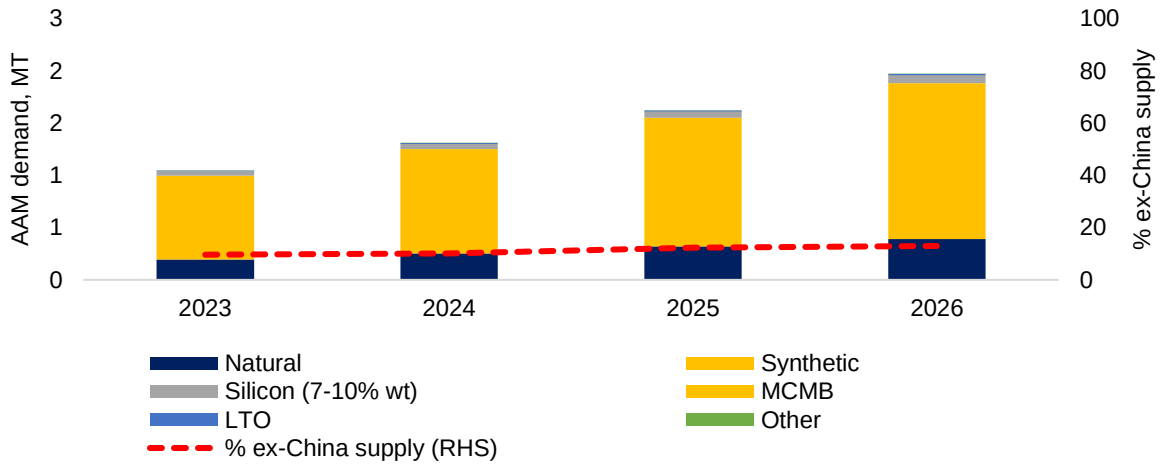
Meanwhile, to produce synthetic graphite AAM, either needle coke (produced in a delayed coker from aromatic slurry oil) or petroleum coke undergoes graphitisation, the process of heating the carbon materials to ~2,500–3,000°C in an electric furnace to transform the material into an ordered graphitic structure. Further steps include coating and baking. Unlike cathode active materials, where the largest cost component is raw materials, AAM costs are largely driven by processing, particularly electricity costs for graphitisation. This partly explains why most Chinese graphitisation operations take place in Inner Mongolia—a source of low-cost electricity.

While natural graphite AAM has slightly higher specific capacity, lower costs, and marginally better environmental standards than synthetic material, around 70–80 per cent of today's global battery anode market is made up of synthetic AAM. This is largely due to the better product quality of synthetic material and its homogeneity. Inconsistencies in natural anode quality have been a bugbear for auto OEMs, who undertake strict qualification cycles to avoid recall risks and cell failure.

Chinese dominance

In no other part of the global battery value chain does China have such dominance as it does in the anode segment. In the natural market, while Africa (Madagascar/Mozambique) accounts for ~15–20 per cent of flake supply, more than 94 per cent of this supply is exported to China for processing/spheronisation. In the synthetic market, while China relies on premium needle coke imports from producers such as Phillips 66 to produce synthetic graphite for high-quality electrodes, it currently hosts around 98 per cent of the world's graphitisation capacity. For both synthetic and natural anodes, China holds around 90 per cent anode market share (see Figure 1), with the biggest ex-China natural anode producer being Posco Chemicals, a key anchor client for ex-China flake producers.

Figure 1: Global battery AAM demand by chemistry and % of ex-China AAM supply



Source: Benchmark Minerals
Note: LTO = Lithium titanate, MCMB = MesoCarbon MicroBeads, RHS = Right Hand Side, Wt = weight

That is not the end of the story, however. Within China itself, anode market competition is one of the toughest. While there are around 75–100 anode companies operating in today's market (80–90 per cent of which are based in China), three Chinese majors - BTR, Shanshan, and Putailai - account for almost 50 per cent of global market share.

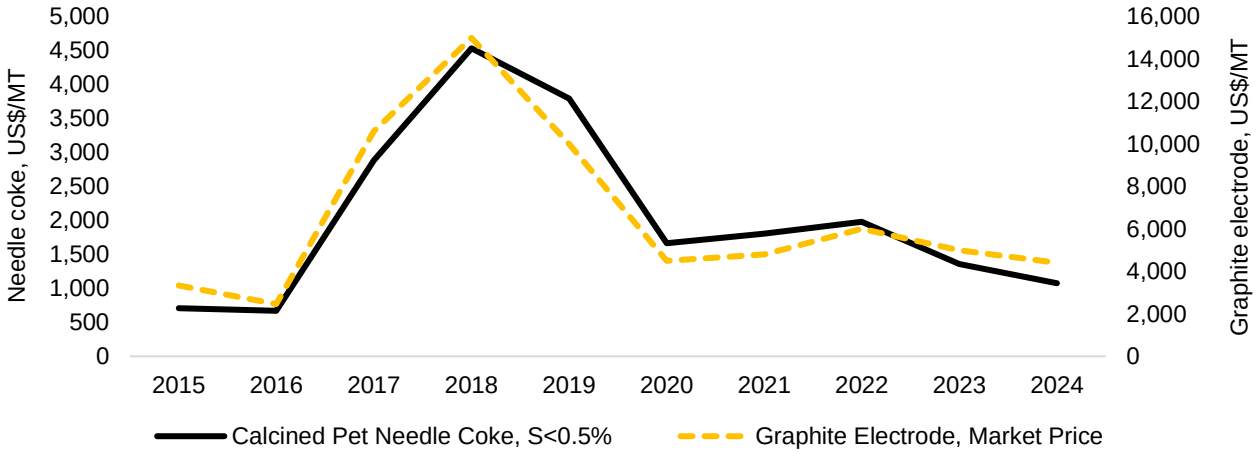
Outside these tier-1 players, the industry is highly fragmented among tier-2 and tier-3 players all competing for cell and OEM customers in what has become one of the most dog-eat-dog markets in the Chinese battery value chain. In this sense, Chinese market behaviour is just as much about domestic competition as it is about preventing new market entrants from outside China.

Cost control and the synthetic pivot

It is not only the consolidated market structure of the anode market that matters for the West. Another key trend over the past few years has been the ability of Chinese anode producers to lower costs.

The graphite market has historically been driven by developments in the steel industry. Synthetic graphite - the feedstock used for electrodes in steel - gained major attention in 2017-18 when a global shortage in needle coke sent prices skyrocketing to nearly US$5,000/t because of China's booming EAF demand (see Figure 2).

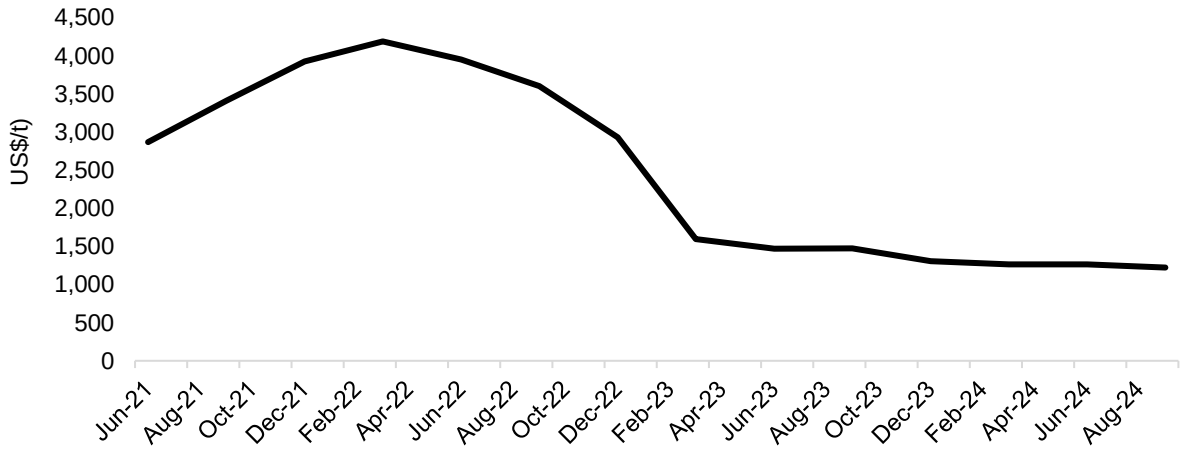
Figure 2: Needle coke and graphite electrode pricing, 2015–2024



Source: Benchmark Minerals, Global Graphite Advisory

While the market corrected shortly after, the next major cycle took place after Covid, for entirely different reasons - namely, the surge in post-Covid battery cell utilisation rates, which boosted demand for battery AAM. Given that graphitisation makes up the bulk of anode production costs (40–50 per cent), non-integrated anode players faced significant margin pressure from the dramatic increase in graphitisation tolling fees (see Figure 3).

Figure 3: Graphitisation tolling fees, 2021–2024

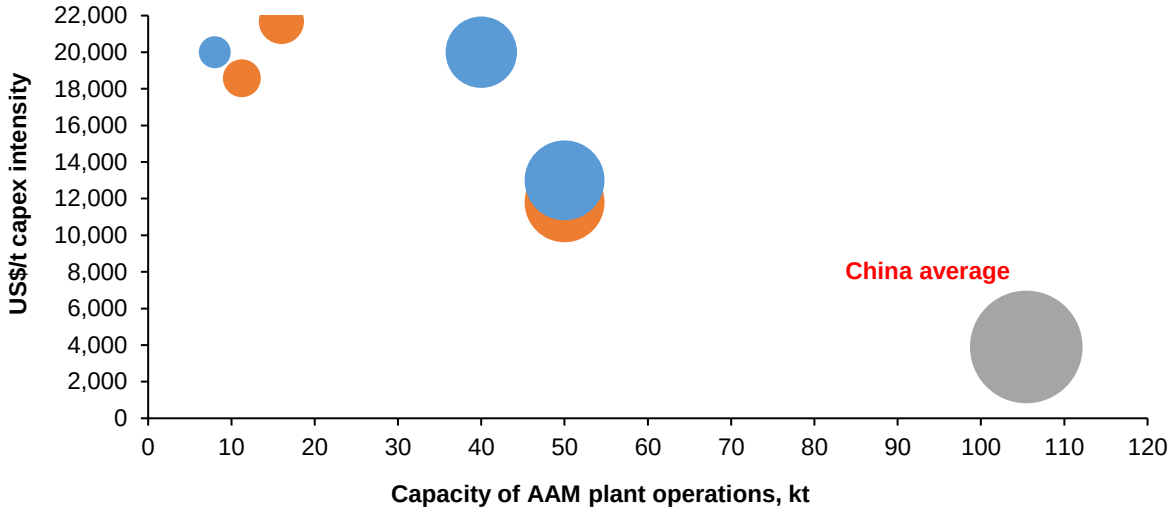


Source: Global Graphite Advisory, BAIINFO

The episode was a major lesson for Chinese anode players. In response, the key cost-cutting exercise executed since 2021–2022 has been greater vertical integration and graphitisation in-housing, reducing the need to use third-party tollers. The second major cost-cutting strategy has been feedstock switching. Chinese anode players have been able to take advantage of the slowdown in China's steel and aluminium industry since 2022 by directing cheaper coke feedstocks into the battery pool. The slump in Chinese aluminium demand in 2022 blew out coke spreads, making it more economic to produce synthetic anode material from pre-calcined cokes, typically earmarked for the aluminium sector. At the same time, the slowdown in Chinese steel demand from the property crisis provided graphite electrode players an opportunity to pivot into the battery market - an easy move given the sunk costs in graphitisation technology.

All three of these dynamics, alongside government subsidies, have helped China significantly increase its synthetic anode capacity, driving a major oversupply into the market.

Figure 4: Capital intensity of selected planed Western AAM projects v China average



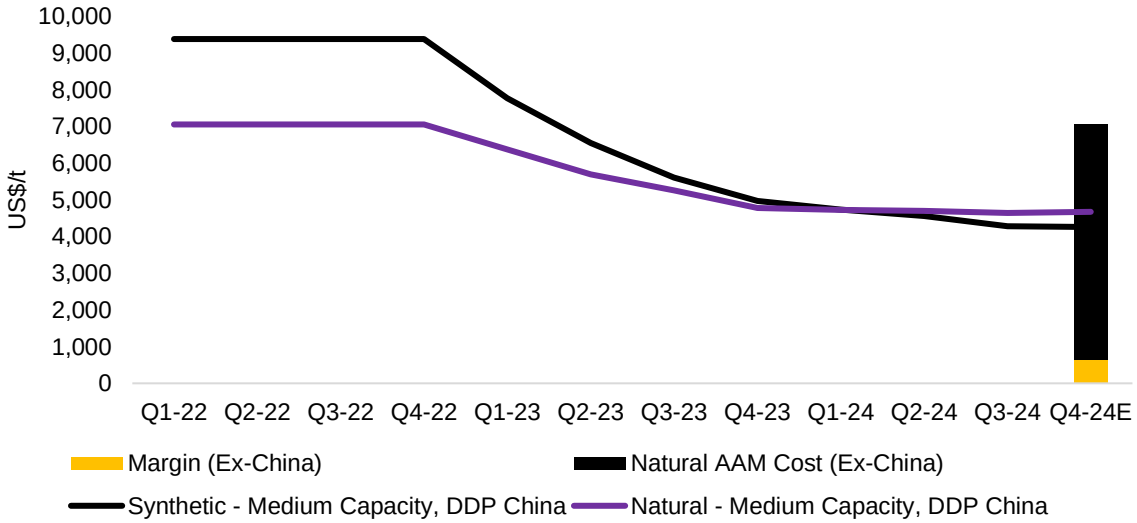
Natural AAM Synthetic AAM

Source: Benchmark Minerals, companies

The strategic implications of these developments have been two-fold:

- First, China's synthetic pivot has narrowed the price differential between natural and synthetic anode material, effectively reducing natural graphite fines demand and keeping up pressure on spot fines pricing.
- Second, in today's ex-China battery anode market, around 65–70 per cent is made up of natural anode players (led by Posco in South Korea), with the rest made up of synthetic anode players. China's synthetic oversupply has effectively compressed the margins of natural players. As shown in Figure 5, Chinese anode prices are now below cash costs of production for ex-China players, effectively putting Western players out of the money.

Figure 5: Synthetic v natural AAM pricing and average ex-China natural AAM break-even price



Source: Author, Benchmark Minerals

Note: DDP = Delivered Duty Paid

Graphite geopolitics

Beyond market structure, graphite geopolitics have also dominated attention in Western OEM boardrooms. In late 2023, China's government tightened export controls on graphite products. While synthetic graphite had already been subject to controls since the mid-2000s, the new policy in 2023 targeted high-purity anode materials, including natural products.

Initially, ex-China anode equities rallied on the news. Panic buying ensued. Korea's graphite industry, seeing the licensing restriction as part of China's retaliation for Western industrial policy, found itself caught between both superpowers, with one Korean trade executive stating: 'we are like a shrimp getting caught in a fight between whales.'[18]

China's move was relevant in the following ways:

- Prior to the move, exports were a way for tier-2 and tier-3 Chinese producers to boost their profitability and offset margin weakness at home. Against this backdrop, tier-1 producers lobbied for a tightening of the regime to cut off this mechanism, helping accelerate rationalisation in the oversupplied Chinese anode market.
- While some western players used the news to bolster their business cases in front of government funding agencies, the reality is that by targeting natural products, China has used the new licensing regime to slowly cut its purchases of western flake supply. In 2024, for example, Syrah Resources, a darling of the Western anode market, reported zero flake material sales to China, amplifying the company's financial pressures.

What can the West do?

Where does this all leave the West? For Western anode players, 2024 has been a mixed year. Despite ongoing price pressures, Western industrial policy—primarily in the US—is helping even the playing field for struggling producers.

In North America, demand-side credits such as the Clean Vehicle Credit (CVC) offer a subsidy of US$7,500 for EV buyers, with the credits linked to OEMs needing to satisfy certain thresholds on critical mineral and component sourcing. Importantly, access to the CVC credits require that no material originates from foreign entities of concern (FEOC) countries, key among them China. While graphite holds a lower value than lithium, for example, the immaturity of the ex-China anode value chain triggered a wave of OEM lobbying to push for delays to restrictions on graphite sourcing. As a result, FEOC guidelines were amended this year to allow Chinese supply to continue uninterrupted until 2027.

While this delay was a blow to ex-China anode players, the Inflation Reduction Act (IRA) CVC credits are a game-changer and a key source of support as Western players launch fresh financing rounds this year and next. Assuming an average pack size in the US EV market of 75 Kwh, US$7,500 would translate to a government subsidy of US$100/Kwh, almost the entire cost of a high-energy-density cell (2024 prices).

In this light, even if US anode producers sign offtakes with cellmakers or OEMs using a floor price well above Chinese spot prices, IRA credit provisions more than offset the price differential.

It is therefore no surprise that 2024 has seen a pickup in offtakes between US OEMs/cellmakers and anode producers, the most notable this year being Panasonic/Novonix, Westwater/SK On, and Nouveau Monde/GM.

While it is possible that the two-year graphite exemption could be extended even further (or that EV credits are removed altogether by President-elect Donald Trump), anode players have also lobbied successfully for protectionist support in the form of tariffs. Historically, tariffs on natural and synthetic AAM were waived in 2020 to 0 per cent, largely a result of Tesla lobbying. However, in May 2024 the US Department of Energy reinstated tariffs at 25 per cent, and these are due to kick in from the mid-2020s.

While these are positive developments for US anode players, the challenge is now execution. China has mastered the anode value chain, and financing teams comfortable with offtake pricing terms will double down on technology due diligence.

Likewise, the reality is that China is still not out. China will continue to act as a source of baseload supply to US OEMs. On top of that, China has not been standing still in response to Western industrial policy. Tier-1 anode players have expanded aggressively overseas, particularly in IRA-compliant trade destinations, a strategy to gain back-door access to the US market. Key destinations for Chinese anode players include Indonesia, Morocco and Europe—the latter now an open sink for Chinese battery players.

Going forward, two other developments will be interesting to watch and could prove a game-changer for the ex-China market. First, oil companies with downstream refining assets could enter the needle coke market and further push into downstream graphitisation. Oil companies are already transforming lithium supply-side economics. Second, an easy source of additional synthetic anode supply for the battery market could emerge from Western graphite electrode players seeking to pivot into the battery AAM market.

[18] Ryu Jeong, China's graphite export controls, Chosun Daily, 21 Oct 2023:

INDONESIA'S ENERGY TRANSITION PARADOX: POWERING THE EV BATTERY SUPPLY CHAIN WITH COAL

Diwangkara Bagus Nugraha, Stein Kristiansen, Indra Overland

Nickel is a key component of nickel-manganese-cobalt oxide batteries, which dominate global EV battery markets.[19] Despite the growing importance of nickel-free battery chemistries such as lithium-iron-phosphate and sodium-ion, the International Energy Agency estimates nickel demand will double from 3.1 million metric tons per year in 2023 to 6 million metric tons in 2040, mainly driven by demand for nickel-rich batteries for EVs.[20]

This article examines the decarbonisation of the Indonesian nickel supply chain from a political economy perspective, highlighting the tensions between economic interests and environmental concerns. These tensions give rise to an Indonesian energy transition paradox: the ostensible purpose of the nickel-based battery industry is to decarbonise the transportation sector, but the battery industry itself relies on newly constructed coal power plants.

Indonesia's nickel industry—booming and carbon-intensive

Indonesia is the world's largest producer of nickel ore and refined nickel and has come to dominate the upstream segment of global nickel supply chains for EVs. This is the consequence of conscious Indonesian policy choices aimed at capitalising on the country's abundant nickel resources.[21] Indonesia supplied 34 per cent of global nickel ore in 2020, growing to 52 per cent in 2023, while its share of global refined nickel production jumped from 23 per cent in 2020 to 37 per cent in 2023. Although most of the nickel is used to make stainless steel, Indonesia plans to gradually shift its production to the battery-grade nickel products nickel matte and mixed hydroxide precipitate (MHP), in the hope of deriving greater economic value from its resources.[22]

While Indonesia takes the lead on the upstream segment of global EV production, its battery and EV manufacturing industries remain at the infant stage. Indonesia inaugurated its first EV battery manufacturer only in July 2024, located in West Java. China's dominance of the global EV supply chain appears even more difficult for Indonesia to challenge than for the EU or the US.

Indonesia's nickel boom is in the spotlight because of its emissions. Battery cathode production, which includes nickel mining and processing, is responsible for more than 40 per cent of carbon emissions from EV production.[23] This is due to the energy-intensive processes involved in cathode production.

Indonesia's heavy reliance on coal for electricity generation and industrial heat results in highly carbon-intensive nickel production. Most nickel processing facilities in Indonesia get their power from captive, off-grid coal power plants with generation capacities of hundreds of megawatts each.

Indonesia's prioritisation of investment over decarbonisation

The ambition of the Indonesian state under Presidents Joko Widodo and Prabowo has been to capitalise on the country's abundant nickel resources to attract more investment, create jobs and business opportunities, and ultimately establish a vertically integrated domestic EV industry.[24] To achieve this, government policy incentivises and provides special treatment for investment in smelters and captive coal power plants while marginalising the issue of greenhouse gas emissions.

The Indonesian nickel boom was launched in 2020 by restricting the export of unprocessed nickel ore in order to stimulate domestic nickel smelting and investment in EV manufacturing. Although this policy was challenged by the EU in the World Trade Organization, Indonesia insisted on continuing to ban exports of unprocessed nickel ore. Following the ban, Indonesia's nickel export value grew by 34 per cent between 2019 and 2023, with a growing share of nickel matte and MHP in exports following the opening of several hydrometallurgical smelters after 2021.[25]

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[19] International Energy Agency (2023), *Global EV Outlook 2023: Catching Up with Climate Ambitions*, Paris: IEA.
[20] International Energy Agency (2024), *Global Critical Minerals Outlook 2024*, Paris: IEA.
[21] Nangoy, F. (2023, 6 February), 'Like Musk, nickel-rich Indonesia has high electric vehicle ambitions', Reuters, https://www.reuters.com/markets/commodities/like-musk-nickel-rich-indonesia-has-high-electric-vehicle-ambitions-2023-02-06/.
[22] ESDM (2023), *Pemerintah Rencana Batasi Izin Pembangunan Smelter Nikel Kelas II, Kementerian Energi dan Sumber Daya Mineral*, https://www.esdm.go.id/id/media-center/arsip-berita/pemerintah-rencana-batasi-izin-pembangunan-smelter-nikel-kelas-ii.
[23] Chen, Q. et al. (2022), 'Investigating carbon footprint and carbon reduction potential using a cradle-to-cradle LCA approach on lithium-ion batteries for electric vehicles in China', *Journal of Cleaner Production* 369, p. 133342, https://doi.org/10.1016/j.jclepro.2022.133342.
[24] Guberman, D., Schreiber, S., and Perry, A. (2024), *Export Restrictions on Minerals and Metals: Indonesia's Export Ban of Nickel*, https://www.usitc.gov/publications/332/working_papers/ermm_indonesia_export_ban_of_nickel.pdf.
[25] Mandala Consulting (2024), *Downstreaming Nickel Industry in Indonesia,* https://mandalaconsulting.id/insights/downstreaming-nickel-industry-in-indonesia.

To incentivise investment in nickel smelters, the government added them to the National Strategic Projects list, rendering them eligible for accelerated permits, eased access to land, and conversion of forests to industrial land.[26] As a result, 54 nickel smelters were operating in Indonesia in 2024, five times the 2020 figure.[27] Most smelters produce class 2 nickel products, such as nickel pig iron and ferronickel for the stainless-steel industry. However, the government plans to halt new class-2 nickel smelter construction and build more battery-grade nickel smelters. In 2024, Indonesia has two high-energy-intensive pyrometallurgical smelters producing nickel matte and five hydrometallurgical smelters, a less energy-intensive smelter type, converting nickel laterite ore to MHP.

China's presence in Indonesia's nickel sector is ubiquitous. As shown in Table 1, most nickel smelters that produce nickel for batteries are backed at least partly by Chinese capital. China is also the main export destination for refined nickel products, as Indonesia has no domestic cathode manufacturers.

Table 1: Battery-oriented nickel smelters in Indonesia

Smelter	Product	Power source	Owners	%	Domicile
Vale Indonesia	Nickel matte	Hydro	PT. Min. Industri Indonesia Vale Sumitomo Metal Mining Publicly traded shares	34.19 34.07 11.55 20.19	Indonesia Canada Japan Indonesia Stock Exchange
Hengjaya Nickel Project	Nickel matte	Coal Solar PV (planned 2025)	Nickel Industries Tsingshan	80.00 20.00	Australia China
Halmahera Persada Lygend	MHP	Coal Solar PV (planned 2025)	Ningbo Lygend Mining Trimegah Bangun Persada Gema Kreasi Perdana	36.90 31.55 31.55	China Indonesia Indonesia
QMB New Energy Materials	MHP	Coal	GEM CATL Tsingshan Holding Group Hanwa Ecopro	63.00 10.00 10.00 8.00 9.00	China China China Japan South Korea
Huayue Nickel Cobalt	MHP	Coal	Huayou Cobalt China Molybdenum Nickel Industries Long Sincere Holding Huaqing Hualon Consulting	57.00 30.00 10.00 2.00 1.00	China China Australia China China
Huake Nickel Industries	Nickel matte	Coal	Huayou Tsingshan Holding Group	70.00 30.00	China China
Huaneng Metal Industry	Nickel matte	Coal	Merdeka Battery Tsingshan Holding Group	60.00 40.00	Indonesia China

Indonesia's president actively persuaded major foreign EV and battery manufacturers to invest in the midstream segment.[28] The state has also taken a direct stake in the battery industry by establishing the Indonesia Battery Corporation, an EV battery manufacturer owned by four state-owned enterprises: Mining Industry Indonesia (a national mining holding company), Aneka Tambang (a national mining corporation), Perusahaan Listrik Negara (the national utility), and Pertamina (the national oil company). To boost domestic demand for EVs and thus for batteries, Indonesia has adopted incentives for domestic EV production. However, so far, EV penetration in Indonesia remains low, with only 2 per cent of new cars sold in 2023 being electric.[29]

[26] Dentons Hanafiah Ponggawa & Partners (2021), *Job Creation Act: Investing and Land Procurement in Indonesian National Strategic Projects Made Easier*, https://dentons.hprplawyers.com/en/insights/alerts/2021/may/19/-/media/dc849da6e6194265bcff621537ea6157.ashx.

[27] Mandala Consulting (2024), *Downstreaming Nickel Industry in Indonesia,* https://mandalaconsulting.id/insights/downstreaming-nickel-industry-in-indonesia.

[28] Cabinet Secretariat of the Republic of Indonesia (2022), 'President Jokowi meets with Hyundai Motor Group chairperson', https://setkab.go.id/en/president-jokowi-meets-with-hyundai-motor-group-chairperson/; Reuters (2022), 'Indonesia's Jokowi meets Tesla's Musk after nickel talks', https://www.reuters.com/world/asia-pacific/indonesias-jokowi-meets-teslas-musk-after-nickel-talks-2022-05-15/.

[29] International Energy Agency (2024), *Global EV Outlook 2024: Moving towards Increased Affordability*, https://iea.blob.core.windows.net/assets/a9e3544b-0b12-4e15-b407-65f5c8ce1b5f/GlobalEVOutlook2024.pdf.

While Indonesia has strong and clear ambitions for its battery-oriented nickel industry, it has shown less interest in the emissions from this industry. The government gives preferential treatment to new captive coal power plants for the nickel industry. Nickel smelters are classified as national strategic projects and thereby excluded from the country's coal power plant moratorium, which was enacted in 2022. As a result, there has been a construction boom for captive coal power plants. In 2023, their capacity to supply the nickel industry reached 7 GW (two-thirds of the country's total captive coal power plant capacity), with an additional 9 GW in the pipeline.[30] As shown in Table 1, six out of seven battery-grade nickel smelters depend on coal power for their electricity supply. Two smelters plan to add solar power in 2025; whether they succeed remains to be seen.

Moreover, Indonesia encourages financing for captive coal power plants for the nickel industry by labelling them amber in the country's Sustainable Finance Taxonomy. As a result, they are eligible for energy transition financing on the condition that they are built by 2031, shut down by 2050, and commit to reducing their carbon emissions by 35 per cent during the first 10 years of operation.[31] Decarbonisation is anticipated to be achieved through electrostatic precipitators, biomass cofiring, or carbon capture and storage, for which Indonesia has high hopes.

Concrete plans for decarbonisation of captive generation were omitted from the 2023 comprehensive investment and policy plan document of the Just Energy Transition Partnership,[32] a G7-led initiative for decarbonising Indonesia's power sector. Apparently, this was due to lack of information about captive generation and the challenge of balancing the government's industrialisation and decarbonisation efforts.[33] The exclusion was expected to be temporary, as the Just Energy Transition Partnership secretariat kicked off a study of captive power for the 2024 plan update.[34]

Indonesian policymakers and the state-owned monopoly electricity company Perusahaan Listrik Negara are unwilling to phase out coal-dependent electricity generation swiftly; they argue that it needs to keep running to ensure affordable and stable electricity for the public and industry, and that Indonesia is too poor to fund the transition.[35] The low coal price cap at US$70 per tonne, and vested coal industry elite interests,[36] result in unattractive conditions for renewable energy investment in Indonesia's major electricity grid on Java, the country's economic and industrial centre. If anyone wants to establish renewables-based battery or EV manufacturing on Java, they will thus have to build their own on-site renewable power plant. In principle, renewable energy certificates can be purchased; but in practice it is difficult to source off-site renewable energy, as power wheeling and direct power purchase agreements with renewable developers are unavailable. According to the Indonesian Ministry of Investment, the unavailability of clean energy was one reason why Tesla abandoned its plan to invest in the country.[37]

Indonesia's lax carbon regulations perpetuate carbon-intensive industrial activities. A carbon cap and trade scheme was introduced in the Indonesian power sector in 2022, starting with smaller coal power plants. As of 2024, its implementation is limited to grid-connected coal power plants, leaving captive coal power plants and carbon-intensive industries untouched by the regulation. There is no carbon tax for the manufacturing industry either.

Moreover, the pressure to disclose carbon footprints is low. Sustainability reporting is only required for companies listed on the Indonesia Stock Exchange. While international financial institutions are increasingly concerned about environmental, social, and governance disclosure for their investments, only 12 of the 54 nickel smelters currently operating in Indonesia disclose their carbon emissions in their annual sustainability reports. And few of those 12 comply with international standards such as the Global Reporting Initiative in terms of the quality and detail of reporting. Nickel companies with shareholders listed on foreign bourses - such as the Hong Kong Stock Exchange, Australian Stock Exchange, Shanghai Stock Exchange, Shenzhen Stock Exchange, or Tokyo Stock Exchange - are subject to additional pressure to disclose their carbon footprint to fulfil the mandatory sustainability disclosure rules of those stock exchanges.[38]

[30] Parapat, J. and Hasan, K. (2023), *Emerging Captive Coal Power: Dark Clouds on Indonesia's Clean Energy Horizon*, Centre for Research on Energy and Clean Air and Global Energy Monitor.
[31] Otoritas Jasa Keuangan (2024), *Taksonomi untuk Keuangan Berkelanjutan Indonesia*, https://www.ojk.go.id/keuanganberkelanjutan/BE/uploads/siaranpers/files/file_ea4f0b98-2e53-419e-8a24-e9503a40a250-19022024225654.pdf.
[32] JETP Indonesia (2023), *Comprehensive Investment and Policy Plan 2023*, https://jetp-id.org/storage/official-jetp-cipp-2023-vshare_f_en-1700532655.pdf.
[33] Suroyo, G. and Nangoy, F. (2023), 'Exclusive: Indonesia to omit private coal power plants from its JETP investment plan', Reuters, https://www.reuters.com/business/energy/indonesia-omit-private-coal-power-plants-its-jetp-investment-plan-2023-10-30/.
[34] JETP Indonesia (2024), 'JETP kicks off captive power study for CIPP update 2024', https://jetp-id.org/news/jetp-kicks-off-captive-power-study-for-cipp-update-2024.
[35] Apriliyanti, I.D., Nugraha, D.B., Kristiansen, S., and Overland, I. (2024), 'To reform or not reform? Competing energy transition perspectives on Indonesia's monopoly electricity supplier Perusahaan Listrik Negara (PLN)', *Energy Research & Social Science* 118, p. 103797, https://doi.org/10.1016/j.erss.2024.103797.
[36] Ordonez, J.A., Jakob, M., Steckel, J.C., and Fünfgeld, A. (2021), 'Coal, power and coal-powered politics in Indonesia', *Environmental Science & Policy* 123, pp. 44–57, https://doi.org/10.1016/j.envsci.2021.05.007.
[37] Tempo (2024), 'Tesla holds off investment in Indonesia due to coal energy, says minister, https://en.tempo.co/read/1912279/tesla-holds-off-investment-in-indonesia-due-to-coal-energy-says-minister.
[38] Kinch, D. (2022), 'Financiers, end-users press for ESG mine ratings as critical minerals demand grows', S&P Global, https://www.spglobal.com/commodityinsights/en/market-insights/blogs/metals/122222-esg-mine-ratings-environmental-social-governance.

The influence of trade partners

In the global supply chains, trade partners can exercise decarbonisation pressure on each other. In 2023, the EU established a battery regulation to tighten the carbon footprint requirements for batteries produced, sold, or used in the European common market. The regulation requires battery producers, distributors, and importers to declare their carbon footprint for their supply chain starting in 2025, with a carbon footprint threshold to be established in 2028. The regulation will create a new carbon standard for battery manufacturers, which may impact Indonesia and other China-linked battery supply chains, as China provides 34 per cent of the EU's battery supply.[39] Meanwhile, China, the primary destination market for Indonesian nickel, limits its emissions regulation to domestic industry and does not cover Chinese overseas investment.

Policy recommendations

Indonesia's nickel industry needs to start decarbonising if it wants to gain access to more capital and wider markets for its products. In that case, the country should enhance its carbon regulations for captive coal power plants and energy-intensive industries such as nickel smelting, improve carbon emission transparency, and consolidate efforts to assist energy-intensive companies in calculating and reporting their emission levels. To decarbonise EV and battery manufacturing, Indonesia would also need to reduce its support for coal to improve the relative attractiveness of investment in renewable energy,[40] incentivise companies to install on-site renewables, and provide manufacturers with access to off-site clean energy through power wheeling or direct power purchasing agreements with renewable energy developers.

SECURING THE FUTURE: JAPAN'S QUEST FOR CRITICAL MINERALS IN THE SHADOW OF CHINA

Parul Bakshi

From the devastating aftermath of the 2011 Fukushima disaster to the ambitious commitments of the Paris Agreement and the stark realities posed by the recent Russia–Ukraine and Middle Eastern conflicts, Japan's energy landscape has undergone profound transformations, catalysing a national reassessment and underscoring the need for a resilient and diversified energy system.

Japan's policies underline the commitment to both domestic resource development and diversified international supply chains. This dual approach aims to mitigate supply disruption risks amidst escalating global energy rivalries and conflicts over low-carbon technology value chains, which are key to competitiveness, economic development, energy sovereignty, and security. With lithium demand projected to triple by 2025 compared to 2020, alongside significant increases in cobalt and copper requirements,[41] immense pressure is placed on global mineral supply chains critical for the energy transition. These challenges are compounded by the dynamic nature of material criticality, driven by economic, geopolitical, and technological shifts, prompting Japan to continually adapt its strategies to meet evolving technological advancements and shifting market demands.

The pursuit of energy security through critical minerals involves navigating vulnerabilities that can slow the pace of the energy transition. Localised disruptions in mineral supply can have ripple effects throughout the global energy system, underscoring the interconnected nature of these resources. Furthermore, geopolitical tensions and market manipulations in mineral-rich regions can inflate costs and prolong reliance on fossil fuels, contrary to the objectives of the clean energy transition.

Japan's acute awareness of its mineral dependencies, given its limited natural resources and high import reliance, has propelled it to take a proactive stance in securing these vital materials. This is evidenced by the Ministry of Economy, Trade and Industry recognising 35 minerals as critical, illustrating Japan's strategic measures to safeguard its technological and energy advancements.[42]

Japan is navigating complex geopolitical and environmental challenges, and its strategy highlights the deep interconnection between national security and climate goals, positioning Japan's international partnerships and innovative policies as a comprehensive blueprint for the global transition.

[39] Hongpei, Z. (2023), 'Makers of EV batteries face stricter EU regulations', *Global Times*, https://www.globaltimes.cn/page/202308/1296311.shtml.
[40] Vakulchuk, R., Overland, I. and Suryadi, B. (2023), 'ASEAN's energy transition: how to attract more investment in renewable energy', *Energy, Ecology and Environment* 8(1), pp. 1–16, https://doi.org/10.1007/s40974-022-00261-6.
[41] Marc Antoine Eyl-Mazzega and Carole Mathieu (2020), *The European Union and the Energy Transition*, Lecture Notes in Energy, volume 73: 27–46, https://doi.org/10.1007/978-3-030-39066-2.
[42] Minoru Nogimori, 'Restructuring of critical mineral supply chain faces "trilemma"', *JRI Research Journal* 7:2 (2024).

China's strategic leverage in global critical minerals

China stands at the forefront of the global supply chain for critical minerals. It dominates this value chain through strategic policies and export controls that significantly affect global market dynamics, influencing both their availability and pricing. Additionally, China's role as a low-cost supplier of clean energy equipment, such as batteries, helps mitigate some economic challenges associated with the global clean energy transition.[43]

This presents a complex dilemma: on one hand, the global dependency on China arises from both demand dynamics and economic advantages; on the other hand, there is a growing impetus for decoupling and diversification from China to ensure stability and security in supply chains. This tension underscores the strategic challenges nations face as they navigate the intricacies of energy transition, balancing economic efficiencies with the imperative for secure, sustainable supply chains.

Despite possessing only 34 per cent of the world's identified rare earth reserves, China's command over the annual mine output and processing stages is significant, with about 70 per cent of global mine output and 90 per cent of refining capabilities under its control.[44] This extensive influence across all stages of the supply chain not only provides China with substantial leverage over global markets but also intensifies concerns about supply security amid geopolitical tensions. For instance, in August 2023, China imposed export controls on gallium and germanium, crucial for semiconductor manufacturing, with additional restrictions introduced on rare earth elements for motor magnets in October and graphite for batteries in December.[45]

China's mining interests are not confined within its borders but span continents, with significant investments in mining operations for copper and cobalt in the Democratic Republic of the Congo and Zambia, lithium in Australia and Latin America, and nickel in Indonesia.[46] These ventures facilitate the return of a significant portion of these minerals to China for processing and refinement, further centralising control within its borders.

While China currently holds a quasi-monopolistic stance in the critical minerals market, it has been suggested that factors such as reserve depletion, the emergence of new mining sites, and enhanced recycling methods could reduce its market share.[47] However, China's existing capacity in mining and processing continues to play a pivotal role, posing significant geopolitical risks that need to be managed through international cooperation and strategic policymaking.

Reassessing security: Japan's response to the 2010 rare earth crisis

Japan's supply chain vulnerabilities came to the forefront during its 2010 diplomatic tensions with China, following a maritime incident near the Senkaku Islands. This conflict, involving a Chinese fishing trawler and Japanese coast guard vessels, escalated to China imposing an informal ban on the export of rare earth minerals to Japan. The embargo caused significant disruptions, especially in the Japanese automotive sector, which relied heavily on these materials for essential components like magnets. While trade normalised after the resolution of incident, the prices of rare earths surged tenfold within a year,[48] highlighting the precarious nature of Japan's supply chain dependency.

In response, Japan developed a comprehensive strategy aimed at mitigating future risks associated with single-source dependency. A supplemental budget of JPY 100 billion was allocated in October 2010, to reduce its dependency on Chinese rare earths from 90 per cent to more manageable levels. The strategic pillars of this initiative included:

- development of alternative technologies that reduce the use of rare earths;
- promotion of recycling initiatives to recover rare earths from used products;
- strengthening of capacities to invest in and develop rare earth mines globally, particularly in Australia, which emerged as a key partner;
- establishment of stockpiles of critical minerals to buffer against supply disruptions.

[43] Nogimori, 'Restructuring of critical mineral supply chain'.
[44] Philip Andrews-Speed (2023), 'Critical minerals for the low-carbon energy transition: why China matters', *Oxford Energy Forum* 137, pp. 10–13, https://www.oxfordenergy.org/wpcms/wp-content/uploads/2023/08/OEF-137.pdf.
[45] Nogimori, 'Restructuring of critical mineral supply chain'.
[46] Andrews-Speed, 'Critical minerals for the low-carbon energy transition'.
[47] International Renewable Energy Agency (2023), 'Geopolitics of the energy transition: critical materials', https://www.irena.org/Publications/2023/Jul/Geopolitics-of-the-Energy-Transition-Critical-Materials .
[48] Tatsuya Terazawa (2023, 13 October), 'How Japan strengthened its rare earth minerals supply chain', World Economic Forum, https://www.weforum.org/agenda/2023/10/japan-rare-earth-minerals/.

Due to prompt policy actions, Japan has effectively reduced its reliance on Chinese rare earth imports from 90 per cent to 60 per cent since the incident, and halved domestic consumption of these materials by 2023 compared to 2010.[49]

In parallel, reports emerged suggesting that China's decline in rare earth exports to Japan was not targeted but rather stemmed from a broader decision, made two months before the trawler collision, to reduce global rare earth exports by 40 per cent.[50] Additionally, a Japanese Ministry of Finance analysis of Japanese port data showed no uniform drop in imports of Chinese rare earths following the incident.[51] Furthermore, a 2023 assessment based on the import share of critical minerals from China found little evidence of selective targeting of its trade partners.[52]

Nevertheless, the broader implications of China's export restrictions came under international scrutiny in 2012 when the United States, the European Union, and Japan challenged China's practices at the World Trade Organization. The complaint centred on China's restrictive export quotas on rare earths, tungsten, and molybdenum, which were viewed as attempts to manipulate the market and maintain a competitive edge. The World Trade Organization ruled against China, highlighting the global implications of national policies for the supply of critical raw materials and underscoring the need for fair trade practices.

Despite the unfavourable ruling, China defended its policies as necessary to protect its environmental and economic interests, indicating a complex interplay between global trade rules and national policy objectives. Regardless of intent, Chinese actions fostered global mistrust and are viewed as a significant example of its assertiveness, leveraging its geopolitical advantage. It served as a watershed moment for Japan, fundamentally reshaping its approach to securing critical minerals.

Japan's geopolitical manoeuvring in the critical minerals arena

As China's influence in the global critical minerals market grows, Japan is strategically diversifying its supply sources and enhancing both domestic and international production capabilities, thereby reshaping the dynamics of energy geopolitics. This strategic shift not only aims to reduce dependency but also fortifies Japan's energy security and economic stability against geopolitical upheavals.

The Japanese government's proactive approach can be termed the '3D strategy': derisk (energy security), decarbonise (environment), and develop (economy), aligned with its 3E+S strategy, which highlights the pillars of Japanese energy policy—energy security, economic efficiency, and environmental protection, plus safety (added post-Fukushima). Since the mid-2000s, Japan has prioritised the identification of critical minerals and risk mitigation through substantial investments in domestic initiatives and international partnerships. This comprehensive strategy emphasises overseas projects, advancements in recycling and substitution methods, and strategic stockpiling, all designed to ensure a stable supply of critical minerals.

Domestic efforts

The New International Resource Strategy (2020) marked a significant step in augmenting these efforts, setting ambitious goals for emergency stockpiling, supporting domestic private-sector mining and smelting initiatives, and increasing stockpile targets to 60 days and up to 180 days for minerals with higher geopolitical risks.[53]

Japan has also intensified its exploration of submarine deposits of critical minerals, discovering six significant sites off Okinawa between 2013 and 2017.[54] Plans are underway to commence commercial exploitation of seabed minerals within its exclusive economic zone starting in 2028, supported by a budget of JPY 215.8 billion. This includes major projects like the exploration of manganese nodules near Minamitorishima, discovered in 2016, estimated to contain 234 million tons and poised to meet long-term mineral requirements—nickel for 75 years, cobalt for 11 years, dysprosium for 730 years, and yttrium for 780 years.[55] However, the environmental impact of large-scale mining on the ocean floor necessitates careful management and licensing from the International Seabed Authority to mitigate damage and align with global environmental standards.

[49] Terazawa, 'How Japan strengthened its rare earth minerals supply chain'.
[50] Amy King and Shiro Armstrong (2013, 18 August), 'Did China really ban rare earth metals exports to Japan?', East Asia Forum, https://eastasiaforum.org/2013/08/18/did-china-really-ban-rare-earth-metals-exports-to-japan/.
[51] King and Armstrong, 'Did China really ban rare earth metals exports to Japan?'.
[52] Simon Evenett and Johannes Fritz (2023, 19 July), 'Revisiting the China–Japan rare earths dispute of 2010', Centre for Economic Policy Research, https://cepr.org/voxeu/columns/revisiting-china-japan-rare-earths-dispute-2010.
[53] International Renewable Energy Agency, 'Geopolitics of the energy transition'.
[54] Jane Nakano (2021), 'The geopolitics of critical minerals supply chains', https://www.csis.org/analysis/geopolitics-critical-minerals-supply-chains
[55] Brad Glosserman (2024, 25 June), 'We've got to get deep-sea mining right', *Japan Times*, https://www.japantimes.co.jp/commentary/2024/06/25/japan/seabed-deep-sea-mining-japan/.

In response to losing 80 per cent of critical minerals through the export of used EVs, the Japanese government has implemented measures to certify the performance of used batteries and promoting local recycling, thereby ensuring these vital minerals are retained within Japan. The Japan Research Institute estimates that a circular economy model for used electric vehicle batteries could significantly increase market value, reaching JPY 600 billion by 2030 and JPY 8 trillion by 2050.[56] Innovation, a key component of these efforts, is seen particularly in the development of perovskite solar cells that utilise iodine—where Japan holds the second-largest share—positioning it as a viable alternative to China-led silicon-based cells.[57]

Underpinning these strategies, the 2022 Economic Security Promotion Act and the National Security Strategy have prioritised reducing excessive dependence on specific countries and establishing a secure, stable supply of critical goods, including rare earths. These policies aim to strengthen Japan's economic security by ensuring a steady supply of essential materials and supporting sustainable practices in the energy and technology sectors.

Role of Japan Organization for Metals and Energy Security

Central to Japan's strategy is the Japan Organization for Metals and Energy Security (JOGMEC), which spearheads initiatives in exploration, recycling, alternative material development, and strategic stockpiling. With branches in 13 countries, JOGMEC enhances Japan's mineral acquisition by offering financial support through liability and debt guarantees, equity participation, and loans to mitigate risks for Japanese enterprises. It aims to achieve 80 per cent self-sufficiency in base metals like copper and nickel by 2030 through collaborations with foreign firms.[58] Following the end of China's rare earth embargo, JOGMEC orchestrated a pivotal $250 million deal with the Australian rare earth company Lynas, securing a continuous supply of rare earths and reinforcing Lynas's role in the global market, where it now provides 12 per cent of the world's rare earth oxides.[59] As of 2023, Lynas supplied about 90 per cent of Japan's neodymium and praseodymium.[60] Between 2004 and 2020, JOGMEC engaged in over 100 projects with investments exceeding $600 million; it currently manages more than 30 projects across 15 countries.[61]

JOGMEC illustrates how effective institutional development and policy implementation can enhance resilience in a complex global environment.

International collaborations

According to Japanese government estimates, installing just 10 GW of offshore wind by 2030 would require about 10 per cent of Japan's 2018 copper consumption and 20 per cent of its niobium rare earth consumption[62] - resources that cannot be fully supplied by recycling and substitution. As a result, Japan is increasingly turning to international collaborations to secure the necessary minerals to meet its energy needs.

These efforts include partnerships with resource-rich countries, such as Australia for rare earths and lithium and Canada for cobalt. Such alliances, often underpinned by bilateral agreements, provide Japan with direct access to critical minerals, reducing its dependency on volatile suppliers like China.

In 2023, Japan and the United States signed an agreement focused on securing minerals for clean vehicle batteries, extending collaborations to South Korea to strengthen semiconductor and mineral supply chains.[63] It invested in a research centre to develop production in Vietnam.[64] Japan is also developing semiconductor facilities and nickel supply chains with the Philippines and bolstering supply chains and mining projects with France and the United Kingdom, with plans for joint investments in Africa.[65]

[56] Yomiuri Shimbun (2024, 3 September), 'Japan govt to promote domestic used EV market; aims to retain critical minerals found in batteries', https://japannews.yomiuri.co.jp/business/economy/20240903-208968/.
[57] Japan Agency for Natural Resources and Energy (2024, 16 February), 'Perovskite solar cells: the key to the future expansion of renewable energy in Japan, part 2: efforts toward early public implementation', https://www.enecho.meti.go.jp/en/category/special/article/detail_196.html.
[58] Nogimori, 'Restructuring of critical mineral supply chain'.
[59] Nayan Seth (2024, 15 August), 'How to diversify mineral supply chains—a Japanese agency has lessons for all', New Security Beat (blog post), https://www.newsecuritybeat.org/2024/08/how-to-diversify-mineral-supply-chains-a-japanese-agency-has-lessons-for-all/.
[60] Joseph Rachman (2023, 15 August), 'Japan might have an answer to Chinese rare-earth threats', *Foreign Policy* (blog post), https://foreignpolicy.com/2023/08/15/japan-rare-earth-minerals-green-transition-china-supply-chains/.
[61] Seth, 'How to diversify mineral supply chains'.
[62] Andrew DeWit (2021), 'The IEA's critical minerals report and its implications for Japan', *Rikkyo Economic Review* 75, no. 1.
[63] Jiji Press (2024, 27 June), 'Japan, U.S., S. Korea agree on deepening critical material supply chain cooperation', *Japan News,* https://japannews.yomiuri.co.jp/business/economy/20240627-195081/.
[64] Samantha DeCarlo and Anna Perry (2023), 'Why can't we be friends? friendshoring the REE supply chain' https://www.usitc.gov/publications/332/executive_briefings/ebot_friendshoring_ree.pdf
[65] Nikkei Asia (2024, 4 April), 'Japan, U.S., Philippines to cooperate in clean energy, chips: Kishida', https://asia.nikkei.com/Editor-s-Picks/Interview/Japan-U.S.-Philippines-to-cooperate-in-clean-energy-chips-Kishida; Takeru Tsuzuki (2024, 2 May), 'Japan, France to cooperate over supply of key minerals; nations to share information on mining, processing tech', *Japan News,* https://japannews.yomiuri.co.jp/politics/politics-government/20240502-183764/; Anna Nishino (2024, 4 September), 'Japan and U.K. to jointly invest in critical minerals in Africa', Nikkei Asia, https://asia.nikkei.com/Politics/International-relations/Japan-and-U.K.-to-jointly-invest-in-critical-minerals-in-Africa.

Additionally, Japan is active in multilateral initiatives such as the Minerals Security Partnership and the Quad Working Group on Critical Minerals, which focus on securing stable mineral supplies for regional security and economic prosperity. Also, during the G7 meeting in 2023, which it hosted, Japan committed over JPY 200 billion to enhance its mineral resources and spearheaded the adoption of the Five-Point Plan for Critical Minerals Security, emphasising the need for diversified supply chains to mitigate strategic dependencies, notably against China's dominance.[66] By continuing to enhance its domestic capabilities and strengthen international partnerships, Japan not only aims to secure its energy transition but also to influence global standards and practices in mineral supply chain management.

Conclusion

Japan's strategic management of critical minerals showcases a comprehensive blueprint, highlighting the importance of a multifaceted strategy that addresses policy, innovation, institutional, and diplomatic fronts. By diversifying its sources and reducing reliance on Chinese rare earths, Japan not only enhances its energy security but also contributes to global market stability, by reducing the world's dependency on a single supplier and mitigating the risk of the kind of supply shocks and price volatility that led to the 2010 crises. These measures help prevent similar disruptions from impacting international economic and technological progress by promoting a more distributed and resilient global supply chain.

But despite robust strategies, Japan faces challenges like intense global competition for resources and potential geopolitical tensions in supply regions. Critics of friend-shoring also argue that Japan's strategy may exacerbate economic disparities and increase geopolitical risks by fuelling competition and inflating prices, potentially leading to heightened areas of conflict and economic inequality. Japan must also recognise the importance of integrating labour rights and environmental considerations into its development strategies, ensuring sustainable and equitable practices.

While Japan's approach to securing critical minerals is not without its flaws, the ongoing mistrust of China's role in the global market underscores the need for diversification of sources, presenting opportunities for other nations to play significant roles in the wider global energy transition and, consequently, in the geopolitics of energy. Japan's approach provides valuable insights into balancing economic, environmental, and security goals. It emphasises the need for ongoing innovation, strong international partnerships, and supportive policies that foster sustainable and fair practices.

US–AFRICAN CRITICAL MINERALS COLLABORATION: REDUCING TRADE FRICTIONS WHILE ENHANCING VALUE ADDITION

Brad Simmons

Amidst decarbonisation imperatives, burgeoning demand for critical minerals for the energy transition has solidified its role as a front line in the increasingly fierce US–China industrial competition. The era of industrial rejuvenation envisioned as an outcome of the Biden administration's climate and energy legislative achievements will mean little for supply chain diversification if the United States cannot wrest market dominance from Chinese actors and achieve the diversification needed to preserve its energy security. Though President-elect Trump's campaign rhetoric would indicate radical departures from the Biden administration's energy and climate policy, resilience in critical material supply chains has been a rare area of bipartisan consensus and continuity between alternating party control of the White House.

Data on investments stimulated by the Inflation Reduction Act (IRA) indicate progress toward the Biden administration's industrial vision, but investment in critical minerals has been substantially lower than other manufacturing investments in clean energy technologies since the start of the administration. Critical mineral industrial investments have made up a mere 6 per cent of the $113 billion in clean energy manufacturing investments in the US between 2020 and the second quarter of 2024, according to the Clean Investment Monitor, a joint database provided by the Rhodium Group and Massachusetts Institute of Technology.[67] Meanwhile, battery gigafactory and associated component manufacturing investments have reached nearly $69 billion over that same period.

This emerging industrial capacity gap poses challenges to both US energy security and industrial competitiveness if China is able to retain its dominant market share in mineral supply chains essential to decarbonisation technologies. The Treasury Department

[66] Takeo Kumagai and Meghan Gordon (2023, 26 May), 'Interview: Japan to boost critical minerals security with G7, "like-minded countries"', S&P Global, https://www.spglobal.com/commodityinsights/en/market-insights/latest-news/energy-transition/052623-interview-japan-to-boost-critical-minerals-security-with-g7-like-minded-countries.

[67] Clean Investment Monitor, Rhodium Group and Massachusetts Institute of Technology, https://www.cleaninvestmentmonitor.org/database.

only finalised the rulemaking process for the most consequential IRA tax credit for critical materials production, 45X, on 24 October 2024. The 45X tax credit provides an incentive for the expansion of advanced manufacturing facilities in the United States. It covers a wide range of clean energy technologies, including batteries, their components, and the production of underlying input materials.

The original proposed rule excluded the extraction of minerals from the costs of production as part of the deductible value of critical mineral processing or component manufacturing. The Treasury Department ultimately reversed this position and included the value of input materials in the deductible, significantly expanding the projected value of the credit.[68] The credit also applies to domestic mining if the miner converts the mined material into a qualifying product, which rewards the type of vertical integration that has contributed to China's dominance of critical material supply chains. Given that the costs of input raw materials represent the vast majority of overall costs in critical material processing,[69] and that these inputs have been subjected to significant price volatility in the recent past, capturing them in the final rule vastly improves the incentive's attractiveness.

The 45X revision—which should now more effectively capture the cost structure of critical material processing and extraction—increases the prospects for domestic greenfield processing and refining capacity for minerals such as copper, lithium, nickel, platinum group metals, and rare earth elements where there is existing domestic production or where project pipelines suggest near-term prospects for initiating or scaling production. Even with new reserve estimates indicating extensive lithium deposits in the United States,[70] domestic demand for critical materials will substantially outpace production in all mineral categories in the near term. That supply–demand imbalance will thus limit the competitiveness benefits that 45X can provide via vertical integration and the efficiency gains of colocation of mining and processing operations. Though the IRA will likely undergo some modicum of reform under a Republican-led Congress and White House, 45x's technology agnostic design and, with its implementing guidance evolution, prospect for enhancing domestic mining, should be a candidate for preservation in alignment with Trump administration objectives.

To effectively prosecute a strategy on critical material supply chain diversification and resiliency, therefore, the United States will require complementary international initiatives targeting energy transition minerals and materials as well as deepening ties with governments that oversee their development. But such initiatives should be crafted carefully, as the risk of overemphasising domestic processing compared to general diversification of supply chains could prove counterproductive if it stimulates excessive protectionism in foreign mineral-rich jurisdictions. There are already indications that this sentiment is hardening: Indonesia's explosive nickel production and smelting growth in the wake of its institution of a raw material export ban is leading other governments in developing countries with substantial resources to consider similarly draconian policy measures. Numerous African governments have referenced Indonesia as a potential model worthy of examination or have quickly followed suit with their own export bans. Zimbabwe and Namibia each enacted bans on the export of raw lithium ore in the past two years, while Ghana, Gabon, South Africa, Zambia, and Zimbabwe have each imposed export restrictions of varying severity and structure in the past decade.[71]

It is clear that export bans can, in certain instances, add value to the mining industry and domestic economy, but even Indonesia's own volatile history of such bans across various commodity classes indicates a decidedly mixed track record. A more nuanced approach by the United States in the wake of the decision to expand 45X's reach to input material costs should therefore be leveraged to drive deeper collaboration with existing and potential critical material producers across the African continent.

This concerted approach to diversifying processing capacity and expanding raw material supply would benefit from existing international platforms that the United States has launched in the past four years, specifically the Partnership for Global Infrastructure Investment and the Minerals Security Partnership (MSP). The former has made commendable progress in devising a collaborative approach with donor countries and development finance institutions to enhancing the enabling environment for the mining industry in central Africa through the formation of the Lobito Corridor.[72] The Corridor's establishment will diversify export

[68] US Department of Treasury (2024, 28 October), 'Advanced Manufacturing Production Credit: A rule by the Internal Revenue Service', https://www.federalregister.gov/documents/2024/10/28/2024-24840/advanced-manufacturing-production-credit.
[69] SAFE (2024, 13 February), 'Letter from SAFE to US Department of Treasury, public comment submission on Docket: REG-107423-23: Section 45X Advanced Manufacturing Production Tax Credit', https://www.regulations.gov/comment/IRS-2023-0063-0133 .
[70] National Minerals Information Center, US Geological Survey (2024, 21 October), 'Unlocking Arkansas' hidden treasure: USGS uses machine learning to show large lithium potential in the Smackover Formation', https://www.usgs.gov/news/national-news-release/unlocking-arkansas-hidden-treasure-usgs-uses-machine-learning-show-large.
[71] Barbara Fliess, Ernst Idsardi, and Riaan Rossouw (2017), *Export Controls and Competitiveness in African Mining and Minerals Processing Industries*, OECD Trade Policy Paper no. 204, OECD Publishing, https://doi.org/10.1787/1fddd828-en.
[72] Lobito Corridor Investment Promotion Authority (2023, 26 October), 'Seven signatories for Lobito corridor extension MoU', https://www.lobitocorridor.org/post/seven-signatories-for-lobito-corridor-extension-mou.

routes for Central Africa's copper belt and is pulling in significant private sector investments to the underlying transport logistics infrastructure to do so. The MSP's origins trace back to the first Trump administration and should be a candidate initiative for continuation. For a region that is frequently undervalued and de-prioritized to the US' detriment, seizing momentum in politically desensitized areas will enable a Trump administration to engender trust and confidence in its African counterparts rather than perpetuating a tired cycle of fresh approaches.

The United States should expand on growing goodwill to help marshal the necessary resources to improve the mapping of Africa's critical mineral resources and create incentives for establishing processing capacity that meets the IRA's unique battery material sourcing requirements,[73] diversify supply chains in segments of severe concentration, and drive further clean technology innovation. A recent International Monetary Fund report found that sub-Saharan Africa possesses 30 per cent of the world's proven critical mineral reserves but attracted only '13 per cent of announced global greenfield foreign direct investment projects in metals and minerals annually'.[74]

The United States should focus on two specific policy initiatives to maximise the impact of 45X on energy transition critical material supply chain diversification. The first would be to partner with Morrocco to institute African sourcing incentives for critical material inputs for its burgeoning lithium-ion battery manufacturing sector. The IRA's EV tax incentive requires that battery critical materials be either mined or processed domestically or in free trade jurisdictions for the vehicle to qualify for the credit. Morocco is the sole African nation that possesses a free trade agreement with the United States, which has precipitated a growing volume of announcements by battery original equipment manufacturers (OEMs) and material producers to establish capacity in Morocco to take advantage of its trade status. Furthermore, Morocco possesses vast phosphate reserves—it was the second largest global producer in 2023 and has an estimated two-thirds of remaining global reserves[75] - that can be utilised in lithium iron phosphate cathode chemistries in batteries. Lithium iron phosphate cathodes have steadily improved their performance characteristics and, given cost competitiveness, have rapidly gained market share from nickel-manganese-cobalt heavy cathode chemistries.

A focused initiative around Morocco has the added benefit of relieving the eligibility bottlenecks created by the IRA's EV tax credit given the lack of nickel and cobalt producing and processing jurisdictions that are also free-trade-agreement partners of the United States. With substantial natural graphite reserves in eastern Africa, the United States and Morocco could also work to address severe concentration in the battery anode and underlying active material input segment of battery supply chains by partnering with producers to develop processing capacity with corporate structuring or equity arrangements that benefit the raw material producing country and its domestic industry.

Announcements of project participants in Moroccan battery precursor material plants indicate a dedicated effort to comply with the IRA EV tax credit's 'foreign entity of concern' sourcing restrictions. Joint ventures between Chinese and Western battery material OEMs, such as LG Chem and Huayou Group[76] and Falcon Energy Materials and Hensen Graphite & Carbon Corporation[77] in Jorf Lasfar, all appear to be geared toward potential IRA-compliant supply to the United States. This catalytic reaction to the IRA is demonstrating the efficacy of its tools in driving geographic supply chain diversification and could improve the capabilities of non-Chinese firms for corporate diversification purposes via these partnerships. As Northvolt's recent struggles indicate, technical and operational challenges surrounding product yield have been a significant constraint in the emergence of home-grown Western competition to Chinese incumbent producers.[78] Leveraging China's learnings and process efficiencies through these partnerships that accord with the IRA's sourcing requirements will be an important evolution in establishment of a more competitive landscape. Before committing to a potential repeal of the IRA's EV tax credit, the Trump administration should consider this collateral damage and the ensuing empowering impact such a decision would have on Chinese EV and battery producers.

[73] See Brad Simmons (2024, May), 'US efforts to decarbonize and de-risk the battery supply chain: are they fundamentally incompatible?', Oxford Institute for Energy Studies, https://www.oxfordenergy.org/publications/us-efforts-to-decarbonize-and-de-risk-the-battery-supply-chains-are-they-fundamentally-incompatible/.
[74] Wenjie Chen et al. (2024), 'Digging for opportunity: harnessing sub-Saharan Africa's wealth in critical minerals', Regional Economic Outlook Analytical Note, International Monetary Fund, https://www.imf.org/-/media/Files/Publications/REO/AFR/2024/April/English/MineralsNote.ashx.
[75] National Minerals Information Center, US Geological Survey (n.d.), 'Phosphate rock statistics and information', https://www.usgs.gov/centers/national-minerals-information-center/phosphate-rock-statistics-and-information.
[76] Heekyong Yang, "LG Chem with China's Huayou to make battery materials in Indonesia, Morocco" Reuters, Sep. 24, 2023, https://www.reuters.com/technology/lg-chem-partners-with-huayou-group-build-joint-lfp-cathode-plant-morocco-2023-09-24/.
[77] Safaa Kasraoui, "Falcon Energy Materials, Hensen team up to build anode plant in Morocco," *Morocco World News,* Sep. 10, 2024, https://www.moroccoworldnews.com/2024/09/365101/falcon-energy-materials-hensen-team-up-to-build-anode-plant-in-morocco.
[78] Yusuf Khan and Claire Brown (2024, 4 October), 'Is Europe's biggest battery maker about to run out of juice?' *Wall Street Journal,* https://www.wsj.com/articles/is-europes-biggest-battery-maker-about-to-run-out-of-juice-f991d280.

The second initiative should focus on the formation of a battery materials and components innovation strategic partnership between the United States, EU, and African Union. As the referenced struggles of Northvolt indicate, a Western battery supply chain diversification strategy will face steep, potentially insurmountable challenges if it focuses purely on existing battery technology. With historic investments in innovation within the battery industrial ecosystem taking place in the transatlantic community, a crowding in of African critical mineral producers to have a stake in that innovation's potential success would help enhance the security of supply of key battery materials. Just as end-of-supply-chain automakers are taking stakes in raw material extraction, so should African financial institutions pursue equity stakes in the next generation of battery producers to diversify their countries' offtake prospects. Such an approach would provide the MSP a more tailored and compelling value proposition for African producers that avoids the stigmas of colonialist tendencies in the continent's extractive industries' past. The incentive to participate in the battery industry's innovation continuum would also address the challenge of shallow African capital markets in the critical materials sector.

Finally, a shared risk for the uptake of 45X for battery component and material producers and immature African mining jurisdictions is the established practice by Chinese incumbents of surging supplies of critical materials to put downward pressure on prices that drive foreign competition out of business. With billions of US taxpayer dollars at stake and the potential of African critical mineral deposits to provide transformative economic development and stability, there will need to be an effective toolkit in place to mitigate this risk. The MSP's recent directive to its development finance and export credit agencies to support critical mineral projects[79] should include a package of tools to combat Chinese market manipulation, including low-interest or concessional bridge loans geared toward injecting liquidity to otherwise financially solvent producers requiring support during periods of artificial market stress. Syndication of such financial tools to include African and US private financial institutions could further stimulate private capital instruments to ensure the durability of the West's response to China's coercive use of its market power.

The reformed approach to the 45X implementation guidelines is a welcomed development for the prospects of critical material onshoring in the United States, but its ability to have a more global impact will require the establishment of complementary, targeted international initiatives. Given African resource-rich countries' significant vested interest in enhancing the value of their reserves, the United States has an opportunity to shape a collaborative approach on further supply chain diversification and resilience.

THE GEOPOLITICS AND CHALLENGES OF MINING IN AUSTRALIA

Ian Satchwell and John Coyne

During the last five years, the significance of concentrated supply chains for critical minerals has surged into the spotlight, primarily driven by geopolitical tensions rather than previously recognised risks. For decades, the world underestimated the implications of China's growing dominance of the global critical minerals market. It did not heed Beijing's use of economic coercion to distort market dynamics and undermine emerging competitors.

As nations scramble to diversify their mineral sources, Australia finds itself at the epicentre of this competition, navigating the dual challenges of being a premier mineral supplier and a target of market manipulation. The unfolding landscape calls for a strategic reassessment of how Australia can leverage its resources and its investment footprint while safeguarding its interests in an increasingly complex geopolitical environment.

A sharpened focus on critical minerals

The term 'critical minerals' originated in the context of resource management and national security, particularly as governments and industries began to recognise the strategic importance of certain minerals for economic stability and technological advancement. In the US, the concept gained prominence in the 1970s during discussions about energy security and supply chain vulnerabilities, particularly in response to geopolitical events like the oil crisis.

In the early 2000s, as global demand for advanced technologies and renewable energy sources increased, the term was formalised in various policy discussions. Governments and organisations, including the US Geological Survey, US Department of Commerce, Japan, and the European Union, began identifying specific minerals as critical based on economic importance, supply risk, and lack of viable substitutes. This recognition has continued to evolve, especially in light of shifting geopolitical dynamics and technological innovation.

[79] Jaime Smyth et al. (2024, 23 September), 'Western nations join forces to break China's grip on critical minerals', *Financial Times,* https://www.ft.com/content/2984ae03-df15-420b-89cc-9ad8337014a9.

During the past two decades, China has built dominant supply chains for several minerals that are key to decarbonising energy systems and high-technology applications, including defence equipment. This dominance resulted from Beijing's strategic foresight and ability to align its public and private sectors, and Western liberal democracies' faith in economic neoliberalism. Today, China's position as a monopoly supplier for some minerals and a near-monopsony buyer for several others gives it extraordinary market power, which it has demonstrated through multichannel economic coercion to control markets and keep out new suppliers.

China is the largest processor of several critical minerals, including copper, lithium, and rare earth elements. In the case of the latter, it has several times withheld supply to customer nations, disrupting the manufacturing processes that use them.

Supply chain security—a strategic imperative

For years, the rest of the world did not pay attention to China's growing mineral market power, but it is now seized by the realisation of how dependent both producer and consumer nations have become on Chinese-controlled supply chains for key minerals and the heightened risks that result.

While the dominance of Chinese control over supply chains is a clear concern, other threats emerge, such as natural disasters disrupting mineral shipments, terrorist attacks on transportation, and community unrest leading to mine closures. Furthermore, inadequate environmental, social, and governance practices within certain supply chains can pose long-term risks. Minerals supply chains encompass all activities involved in exploration, mine development, mineral production and processing, and recycling of end products. Typically crossing multiple borders, these supply chains are more resilient and secure when diverse. In contrast, concentrated chains face higher risks of disruption from single points of failure.

Australia is both a world-leading mineral producer and one of the world's largest investors in mining globally. This inevitably brings it into the orbit of China, Australia's and the world's largest minerals customer and a growing investor in minerals supply chains.

As major manufacturing nations seek to reduce supply risk by diversifying mineral supply chains away from China and some other dominant suppliers, Australia now finds itself at the centre of global investment and supply chain competition. This goes beyond simply where its mineral investment comes from and where its mineral products are sold. Mineral supply chains are now intertwined with strategic alignments and security.

Therefore, how Australia responds will be key to its export-driven economy, the ongoing success of its minerals companies operating around the world, and its security and that of its partners.

Disruption of efforts to diversify supply chains

While Australia's competitive advantages in minerals include its reliability as a supplier, it is not immune to interruptions in its mining and shipping, which consequently affect global supply. The market price of manganese, a key ingredient in steel alloys, spiked in 2024 in response to lengthy interruptions of shipments from a large mine in northern Australia due to cyclone damage to a loading jetty.

Despite concerted action by Japan, South Korea, the US, Canada, and Australia, only one Australian-based rare earths project, the Lynas Rare Earths Mount Weld mine and processing facilities,[80] have been established and expanded. Several other rare earths projects are struggling to be commercially sanctioned in the face of depressed market prices, likely influenced by China's economic coercion, despite financing support from Australia and partner nations. Even though China is highly dependent on Australia for its iron ore and, latterly, lithium supplies, that has not stopped it from using economic coercion to control new rare earths production from Australia and keep out would-be new Australian market entrants. The long-established mineral sands company Iluka Resources, which is seeking to build a heavy rare earths refinery in Western Australia, has been very vocal about how China's market manipulation of these strategically crucial products has affected project economics.

Attempts by China-linked shareholders to gain control of a proposed rare earths project, Browns Range, in northern Australia triggered orders by the Australian government in 2024 for a China-linked shareholding sell-down on the recommendation of the government's Foreign Investment Review Board.[81]

China's massive investments in low-cost but high-impact nickel mining and processing in Indonesia have created a new supply stream for battery manufacture and stainless steel. Consequently, depressed global nickel prices have forced Australian production out of the market, reducing global supply diversity and shutting down environmentally responsible mining and processing operations.

[80] Lynas (2024), website https://lynasrareearths.com/, Lynas Rare Earths Limited, Perth

[81] Federal Register of Legislation (2024), Foreign Acquisitions and Takeovers (Disposal of Interests in Northern Minerals Limited) Orders 2024, https://www.legislation.gov.au/F2024N00475/asmade/text. Australian Government, Canberra.

Meanwhile, in Africa, China's economic and geopolitical coercion has led to the sale of two Australian-operated lithium mines, in Mali [82] and the Democratic Republic of the Congo [83], to Chinese hands. In both cases, apparently trumped-up disputes between the mining companies and host governments led to effective expropriation of the assets, with Chinese companies then assuming control.

Australia's challenge to balance relationships

The challenge for Australia often lies in balancing its relationships with China, its largest trading partner, and the United States, its key security ally. While China dominates trade volumes, the US plays a crucial role in Australia's economy, particularly when considering two-way investment and the technology and knowledge exchanged. Additionally, the US is Australia's most important strategic partner, with their collaboration deepening through initiatives like the AUKUS (Australia, UK, and US) alliance and the Quadrilateral Security Alliance (the Quad). This dynamic makes Australia's position increasingly complex and multifaceted.

In mining, however, China remains not only the largest customer for Australian minerals but also an important investor in Australia through equity, joint ventures, and offtake and financing agreements. Direct investment in minerals and energy by China-resident companies between 2006 and 2023 totalled US$47 billion.[84] While year-on-year investment flows have fallen as China diversifies its minerals and energy investments to other destinations, many Australian mines and processing plants owe their existence to the involvement of Chinese companies. Therefore, the Australian government must walk a careful line between China's importance as a customer and investor, and Australia's participation in developing new minerals supply chains with like-minded nations.

Over the past two years, these nations, led by the US, have implemented a flurry of agreements and strategies designed to diversify and secure their critical minerals supply chains. But it is sobering to realise that it was only in 2020 that Washington waved through the sale of the US-owned Kisanfu copper-cobalt mine in the Democratic Republic of the Congo to a partially state-owned Chinese company.

Australia's centrality to critical minerals supply chains is evidenced by it becoming a party to 27 bilateral and multilateral agreements and processes with like-minded nations and the European Union.[85] Typically, these agreements entail parties' commitments to strengthen bilateral supply chains with Australia. The Australian government and its subnational governments, which control mining in their jurisdictions, have rolled out new policies and incentives to increase domestic investment in mining and processing and lift critical minerals output.

About half of the agreements signed by Australia also involve parties working with developing nations to help them enter new, sustainable supply chains and garner lasting benefits. This is implicit endorsement of leadership in responsible mining by Australia and its companies around the world. However, how Australia delivers on its commitments to work with other producer nations under these agreements remains to be seen.

Australia's policies on inbound China-linked investment contrast with the much more hard-line stance of the US, which, through its Inflation Reduction Act (IRA)[86] and its Foreign Entities of Concern (FEOC) rule, severely restricts China-linked minerals investment and imports of minerals and products made from them, such as electric vehicles. Canada has placed strong restrictions on China-linked investment in mining there, largely to protect Canadian minerals exports and manufacturers from falling foul of US import restrictions.

European nations' investment and mineral sourcing policies take a pragmatic approach to China-influenced supply chains. Representatives of Germany, France, and the Netherlands indicated to a mining conference in Australia in October 2024 that their new national investment funds for critical minerals were relaxed about investment in projects alongside Chinese companies in well-governed nations like Australia.

[82] Leo Lithium (2024), CEO AGM Presentation Proposed disposal of shares in Mali Lithium BV and interest in the Goulamina Project, https://api.investi.com.au/api/announcements/lll/3275dae1-107.pdf, Leo Lithium Limited, West Perth.
[83] AVZ Minerals (2024), Manono Project Update, https://avzminerals.com.au/investor-centre/2024/9/26/manono-project-update, AVZ Minerals Limited, West Perth.
[84] KPMG (2024), Demystifying Chinese Investment in Australia, https://kpmg.com/au/en/home/insights/2023/12/demystifying-chinese-investment-in-australia.html, KPMG, Sydney.
[85] Satchwell, Ian (2024), Reclaiming leadership: Australia and the global critical minerals race, https://www.aspi.org.au/report/reclaiming-leadership-australia-and-global-critical-minerals-race, Australian Strategic Policy Institute, Canberra
[86] The White House (2023), Building a Clean Energy Economy: A Guidebook to The Inflation Reduction Act's Investments in Clean Energy and Climate Action , https://www.whitehouse.gov/cleanenergy/inflation-reduction-act-guidebook/, The White House, Washington DC

Working with developing nations to build supply chains

The US, EU, and UK have all implemented actions to fund new projects in developing nations and work with host nations to build capacity in infrastructure and minerals governance. The sustainability of minerals supply chains is a primary objective, in addition to enhanced diversity and security.

South Korea is joining them, signing agreements in June 2024 with 23 African nations to cooperate in minerals, energy, manufacturing, and infrastructure.[87] Ensuring a stable supply of critical minerals, collaborating on minerals production technology, increased investment by Korean companies in Africa, and deriving value from mineral production are four explicit objectives. Korean and Australian minerals interests inevitably will closely interact.

The US-auspiced Minerals Security Partnership (MSP)[88] recently added closer international cooperation in project financing to its suite of tools to establish new, more secure, sustainable supply chains. The MSP Finance Network,[89] involving project financing and export credit agencies from multiple countries, seeks to strengthen cooperation in information exchange and co-financing of critical minerals supply chains.

Australia and Canada, through minerals companies headquartered in the two nations, are Africa's two largest minerals investors. Despite recent efforts of like-minded nations to build more diverse critical minerals supply chains from African nations, 2024 data from S&P Global Market Intelligence show that the number of mines owned by China-based companies has increased by 21 per cent since 2019.[90] The number of mines owned by Australian and Canadian interests has fallen by more than 8 per cent. Partly balancing that decline, the number of mines owned by UK companies increased by 11 per cent.

Risks of policy divergence

There is a growing risk that market interventions by like-minded nations will diverge and even work at cross purposes. For example, the US financially supports Australian companies in developing critical minerals supply chains from mines in Australia, Africa, and Latin America, with further processing in the US. In the meantime, the Australian government currently restricts its financial support to critical minerals mining and processing operations within Australia, which in some cases must get raw materials such as graphite from mines overseas. This seemingly ignores Australia's commitment to a global supply chain approach involving other producer nations. Australia's policy may change due to its commitment to co-financing within the new MSP Finance Network.

Mining and processing projects in Australia are also affected by US IRA and FEOC provisions. A large US-owned lithium hydroxide refinery in Western Australia has suspended an already-underway expansion and reduced its output in the face of depressed lithium prices and because it has fallen foul of the FEOC rules due to the source of its lithium raw material being part-owned by a China-linked company. In the meantime, it remains eligible for Australia's 10 per cent production tax incentive for processed critical minerals.

The net result is that the production of sorely needed alternative lithium hydroxide has been curtailed. At the same time, Australia, a key critical minerals partner to the US, loses major investment.

Achieving a green premium to recognise high standards

A major hurdle to building new, secure, and sustainable minerals supply chains is that the standards applied often bring higher costs than minerals produced less responsibly. Minerals markets have so far been unwilling to pay 'green premiums' despite the increasing focus on sustainability. The London Metal Exchange recently reported that there was little interest from mineral buyers in paying premiums for sustainably produced minerals.[91]

[87] Ministry of Foreign Affairs (2024), Joint Declaration 2024 Korea-Africa Summit, https://www.mofa.go.kr/eng/brd/m_5674/view.do?seq=321015, Ministry of Foreign Affairs, Republic of Korea, Seoul

[88] Department of State (2024), Minerals Security Partnership, https://www.state.gov/minerals-security-partnership/ US Department of States, Washington DC

[89] Department of State (2024b), Joint Statement on Establishment of the Minerals Security Partnership Finance Network, https://www.state.gov/joint-statement-on-establishment-of-the-minerals-security-partnership-finance-network/, US Department of States, Washington DC

[90] S&P Global (2024), China, UK expand mining presence in Africa; US, Canada, Australia lose ground, https://www.spglobal.com/marketintelligence/en/news-insights/latest-news-headlines/china-uk-expand-mining-presence-in-africa-us-canada-australia-lose-ground-83154484, S&P Global, New York

[91] LME (2024), General Update: Discovering the Low Carbon Premium for the Nickel Market, https://www.lme.com/api/sitecore/MemberNoticesSearchApi/Download?id=1dcae570-87ee-422a-9596-acaa2b041f11 The London Metal Exchange, London

The first hurdle - defining sustainability standards and implementing governance and assurance processes - is on its way to being cleared. A consortium of mineral industry bodies recently commenced consultation on the first draft of the new Consolidated Mining Standard,[92] bringing together four voluntary mining standards into one global standard accessible to any company with a commitment to mine responsibly. While a standard for mining and processing may fall short of an end-to-end supply chain standard, it will cover the highest-risk parts of supply chains across all the dimensions of sustainability.

Australia's leadership, along with Canada's, in global minerals investment and mining places these countries at the centre of strategic efforts by major manufacturing nations to develop more diverse, secure, and sustainable supply chains for critical minerals, and to decouple from dependency on China. How effectively like-minded nations with differentiated interests work together globally to achieve this will largely determine their success in achieving mineral security goals.

Meeting challenges; supporting supply chains

Australia faces significant challenges in developing its critical mineral supply chains amidst shifting geopolitical conditions, economic coercion, and market distortions. Balancing its need to engage with China, its largest trading partner, against its commitment to the strategic imperatives of the US requires careful navigation of complex interests. To ensure resilience, Australia must broaden its contribution to supply chains, enhance collaboration with like-minded nations, and establish robust sustainability standards that can withstand external pressures. At the same time, in the face of increased minerals competition, Australia must address its inherently high-cost environment through attention to approval timelines, land access, construction and operating costs, and productivity.

As it moves forward, Australia will need both to fortify its mining and processing capabilities and to play a pivotal role in fostering international cooperation, thereby positioning itself as a leader in the responsible production and supply of critical minerals globally.

LATIN AMERICA'S POSITIONING IN CRITICAL MINERAL GEOPOLITICS

Tom Moerenhout and Victoria Barreto Vieira do Prado

Latin America's critical minerals are essential to the global energy transition. How the region decides to manage this wealth can be decisive: it could position itself as an active geopolitical player, or echo the extractivist-export model of the past and fail to capture economic development opportunities linked to its mineral wealth. This challenge drives the Latin American geopolitical quest for a partner, not just an offtaker. Understanding this, and acting on it, will help the US, EU, or China to develop a comparative advantage in their competition with each other.

Latin America's potential to support critical minerals supply chain diversification

Latin America's potential in critical minerals is well known, especially for copper and lithium, of which the region produces about 40 per cent and 35 per cent of world supply, respectively.[93] However, the potential for growth is much larger than its current production. The region produces an amount equivalent to its share of global reserves only for copper, tin, and zinc. For nickel, graphite, lithium, and rare earths, it produces much less than its share of global reserves. That means there are attractive deposits with high ore grades that are still untouched. Brazil, for example, holds 20 per cent of commercially viable rare earths reserves, but only produces 0.02 per cent.[94] Similarly, it holds 12 per cent of global nickel reserves, but only produces about 2.5 per cent.[95]

That growth potential comes with a set of challenges and opportunities. One of the challenges is strong competition from sub-Saharan African countries. In this region, countries' share of global production is often larger than their share of global reserves, but social, environmental, and governance regulations are often less strict. This has at times given the region an edge in terms of investment in extractives, despite its greater political and regulatory risk.

Global markets, however, are changing, and are increasingly requiring better mining standards. This is very much in line with local dynamics in mining regions in Latin America, where conflicts with local communities over mining standards are ever more

[92] ICMM (2024), Consolidated Mining Standard Initiative, https://miningstandardinitiative.org/, ICMM, The Copper Mark, Mining Association of Canada and World Gold Council
[93] USGS, 2024, 'Minerals Commodities Summaries 2024: Rare Earths', https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-rare-earths.pdf
[94] USGS, 2024, 'Minerals Commodities Summaries 2024: Rare Earths', https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-rare-earths.pdf
[95] USGS, 2024, 'Minerals Commodities Summaries 2024: Nickel', https://pubs.usgs.gov/periodicals/mcs2024/mcs2024-nickel.pdf

pronounced. Several experts believe that market bifurcation between China and the West will be accelerated by relying on the identification and marketisation of higher standards.[96] This is definitely where Latin America, while not perfect, has a comparative advantage.

The region's electricity grid is already very low carbon, with a 60 per cent share of renewables and abundant additional low-carbon resources.[97] This will help with decarbonising mining operations, which is a key strategic goal of mining majors. Environmental regulations are also becoming stricter, given waste-related tragedies in the past and water conflict in the present. This incentivises the development of more environmentally friendly mining and waste management practices. Significant unexplored reserves are also welcoming exploration and pre-feasibility studies, which can help uncover new high-grade ore deposits that can be developed with higher environmental, social, and governance (ESG) standards.

One of the bigger challenges for Latin American countries has been their long permitting times, complex fiscal environments, and regulatory and political instability. That said, several countries are actively updating their regulatory frameworks to capture more investment in responsible mining practices. In particular, Chile, Argentina, Brazil, Mexico, and Peru score relatively well on indicators such as mining sector confidence, specialised labour, infrastructure risk, and expropriation risk.[98] This matters a great deal for the industry given the risky, capital-intensive nature of new greenfield projects coupled with a fairly depressed critical minerals market.

Finally, the region is a well-established diversified exporter. Historically it has close trade connections with the West. This includes the EU–Mercosur agreements, as well as bilateral trade agreements between the EU and Chile, Colombia, Peru, and Mexico, and framework agreements with Argentina and Brazil.[99] The US has the United States-Mexico-Canada Agreement and trade agreements with Chile, Peru, and Colombia.[100] That said, the region is tilting ever more towards China, especially for critical minerals, given China's role as the world's minerals processing hub. Chile signed the region's first free trade agreement (FTA) with China in 2005; Peru followed in 2009 and Costa Rica in 2010. China is now actively pursuing trade deals in the region. In 2023 alone, the country signed five major deals including an FTA with Ecuador, 15 trade-related agreements with Brazil, and key agreements with Argentina and Nicaragua.[101]

The relative importance of trading partners for critical minerals is clearly seen in trade data. In 2022, Chile exported US$25 billion of copper to China, but only US$3 billion to the US. Similarly, Peru exported US$12 billion of copper to China but less than US$1 billion to the US. Of course, China is a copper guzzler, given its stage in economic development. But the situation is very similar for lithium, with Chile exporting US$6 billion to China, and US$0.2 billion to the US.[102] A lot of these exports were raw materials, which shows that for the US to benefit from Latin America's potential for diversification, there will need to be a solution (read: investment) in ex-China processing capacity.

Great-power competition and Latin America's quest for a partner, not an offtaker

The establishment of partnerships can enable the region to capture value from investments to support the development of local industry, job creation, and technological advancement. Such investments can also address infrastructure issues and mitigate operational risks which currently delay industrial development in the region, thus positioning Latin America as a more attractive partner as global competition for critical minerals intensifies between the US, the EU, and China.[103]

[96] See for example Henry Lazenby, 2024, 'Push for ESG price premiums may reshape global critical minerals markets', Mining.com, https://www.mining.com/push-for-esg-price-premiums-may-reshape-global-critical-minerals-markets/ ; Elisa Yoshiara, Fabian E. Villalobos, 2024, 'Critical minerals can pave the road to more robust international development', WEF, https://www.weforum.org/stories/2024/01/critical-minerals-international-development/; Kaycee May Royer, 2024, Leveraging ESG Principles to Help Secure Critical Mineral Supply Chains', Public Land & Resources Law Review, https://scholarworks.umt.edu/cgi/viewcontent.cgi?article=1799&context=plrlr.

[97] IEA, 2023, 'Latin America Energy Outlook, 2023', https://www.iea.org/reports/latin-america-energy-outlook-2023/executive-summary

[98] EIU, 2024, 'Critical minerals in Latam : oppportunities and challenges', https://www.eiu.com/n/critical-minerals-in-latam-opportunities-and-challenges/

[99] EU-Latin America Partnership on Raw Materials, 'Mineral Development Network Platform', https://www.mineralplatform.eu/cooperation/international-agreements-frameworks/eu-latam-agreements

[100] Office of the United States Trade Representative, 'Free Trade Agreements', https://ustr.gov/issue-areas/industry-manufacturing/industrial-tariffs/free-trade-agreements#:~:text=U.S.%20and%20Panama%20signed%20the%20United%20States%2DPanama%20Trade%20Promotion%20Agreement%20on%20June%2028%2C%202007.%20Panama%20approved,15%2C%202012

[101] Bruno Hernandez, 2023, 'China-Latin America and the Caribbean : Investment, Trade, and Future Prospects', China Briefing, https://www.china-briefing.com/news/china-latin-america-and-the-caribbean-investment-trade-and-future-prospects/#:~:text=Ever%20since%20China%20and%20Chile%20signed%20the%20very%20first%20trade%20partnership%20in%202005%2C%20China%20has%20focused%20its%20

[102] Center on Global Energy Policy, 2024, 'Critical Materials Monitor : Trade Explorer', https://criticalmaterials.energypolicy.columbia.edu/explore/?m=Li&t=&c=32&f=export

[103] Center on Global Energy Policy, 2024, 'Materials Powering the future of Energy, https://criticalmaterials.energypolicy.columbia.edu/

The US–China rivalry has become a crucial context for Latin America's engagement with these two powers. The region holds strategic influence for key strategic sectors such as batteries and automobiles in the world's two biggest economies. While China's deepening presence in the region is viewed as an opportunity for local economies, the US is also trying to leverage policies such as the Inflation Reduction Act to encourage US supply chain partners to diversify and reduce dependence on Beijing. The competition between these two powers presents Latin American countries with the opportunity to negotiate stronger, more balanced partnerships.[104]

Foreign investors have taken note of Latin America's potential. In 2023, the region raised over $19 billion amongst 19 megaprojects across different sectors, 17 of which are funded by foreign partners. Commodities and critical minerals represented 23 per cent of the region's greenfield project investments in the last two years, twice as much as in other developing regions.[105] Currently, China is the largest buyer of these minerals in the region.[106] Its firms have been beating offtake options from American and Russian companies across the region. For example, Chinese firms have most recently won a bid to develop two lithium projects in Bolivia,[107] bought lithium rights in Argentina,[108] and poured hundreds of millions of dollars into renewable energy projects in the region.[109]

Chinese influence has expanded beyond mining and into EV manufacturing and refining. These projects serve not only to increase Chinese influence over supply chains, but also to tap into other markets, locally and regionally.[110] This is also an active strategy to prove its intention to strengthen its economic ties with the region.[111] Its investments usually have less stringent economic and political conditions than those of Western lenders, making them more attractive for some Latin American countries. However, they also often involve looser environmental standards and higher debt burdens.[112]

Cognisant of growing Chinese influence, the EU and the US have been moving to gain influence in the region. For instance, both the US and EU are forming partnerships with Latin American countries to ensure the security of supply chains, while also encouraging sustainable extraction and local processing.[113] German Chancellor Olaf Scholz recently travelled to Argentina and Chile in search of lithium deals, and President Biden's Americas Partnership for Economic Prosperity promises multi-sector cooperation with the western hemisphere and the region in the realm of critical minerals.[114] Still, it seems the US has not yet fully woken up to the Latin American potential for de-risking its critical mineral industry.[115] US investments, compared to China's, are fairly small, and institutionally a lot can be improved. For example, the Mineral Security Partnership, an alliance led by the US and the EU to secure and diversify supply chains away from China, has 15 partner countries, none of which are in Latin America.[116]

This absence underscores a broader challenge: while the US Free Trade Agreements with Chile, Colombia, and Peru offer a pathway for these countries to access benefits under the Inflation Reduction Act, including tax incentives for green energy supply

[104] Ilaria Mazzocco, Ryan C. Berg, Rubi Bledsoe, 2024, 'Driving Change : how EVs are reshaping China's economic relationship with Latin America', CSIS, https://www.csis.org/analysis/driving-change-how-evs-are-reshaping-chinas-economic-relationship-latin-america
[105] UNCTAD, 'Latin America and the Caribbean : critical minerals drove foreign investment in 2023', https://unctad.org/press-material/latin-america-and-caribbean-critical-minerals-drove-foreign-investment-2023
[106] Christina Lu, Rocio Fabbro, 2023, 'China's Latin America Gold Rush is All About Clean Energy', Foreign Policy, https://foreignpolicy.com/2023/02/27/china-latin-america-lithium-clean-energy-trade-investment/
[107] Daniel Ramos, 2023, 'Bolivia taps Chinese battery giant CATL to help develop lithium riches', Reuters, https://www.reuters.com/technology/bolivia-taps-chinese-battery-giant-catl-help-develop-lithium-riches-2023-01-20/
[108] Reuters, 2022, 'China's Ganfeng Lithium to buy Lithea Inc for $962 million', https://www.reuters.com/markets/deals/chinas-ganfeng-lithium-buy-lithea-inc-962-mln-2022-07-11/; Reuters, 2022, 'China's Zijin Mining to invest $380 mln in Argentina lithium plant', https://www.reuters.com/article/business/chinas-zijin-mining-to-invest-380-mln-in-argentina-lithium-plant-idUSL6N2UF06P/#:~:text=BUENOS%20AIRES%2C%20Feb%204%20(Reuters,of%20Production%20said%20on%20Friday.
[109] Christina Lu, Rocio Fabbro, 2023, 'China's Latin America Gold Rush is All About Clean Energy', Foreign Policy, https://foreignpolicy.com/2023/02/27/china-latin-america-lithium-clean-energy-trade-investment/
[110] Ilaria Mazzocco, Ryan C. Berg, Rubi Bledsoe, 2024, 'Driving Change : how EVs are reshaping China's economic relationship with Latin America', CSIS, https://www.csis.org/analysis/driving-change-how-evs-are-reshaping-chinas-economic-relationship-latin-america
[111] Ilaria Mazzocco, Ryan C. Berg, Rubi Bledsoe, 2024, 'Driving Change : how EVs are reshaping China's economic relationship with Latin America', CSIS, https://www.csis.org/analysis/driving-change-how-evs-are-reshaping-chinas-economic-relationship-latin-america
[112] Christina Lu, Rocio Fabbro, 2023, 'China's Latin America Gold Rush is All About Clean Energy', Foreign Policy, https://foreignpolicy.com/2023/02/27/china-latin-america-lithium-clean-energy-trade-investment/
[113] J.P. Morgan, 2024, 'Global race for critical minerals… a unique opportunity for latin America', https://privatebank.jpmorgan.com/latam/en/insights/markets-and-investing/global-race-for-critical-minerals-a-unique-opportunity-for-latin-america
[114] Christina Lu, Rocio Fabbro, 2023, 'China's Latin America Gold Rush is All About Clean Energy', Foreign Policy, https://foreignpolicy.com/2023/02/27/china-latin-america-lithium-clean-energy-trade-investment/
[115] Christopher Hernandez-Roy, Henry Ziemer, Nathaniel Laske, 'De-risking Critical Mineral Supply Chains: The Role of Latin America', CSIS, https://www.csis.org/analysis/de-risking-critical-mineral-supply-chains-role-latin-america
[116] US Department of State, 'Minerals Security Partnership', https://www.state.gov/minerals-security-partnership/#Principles

chains, a significant share of Latin America's minerals still flows to China for processing.[117] For instance, China consumed over 65 per cent of Chilean mineral exports in 2021, valued at $22.5 billion (roughly 6 per cent of Chilean GDP).[118] Leveraging these FTAs more effectively could help the US pivot to support processing not only in the US but also in Latin America, so that the potential to diversify from China and support Latin America's offtaker diversification is maximised.

China's strategy, on the other hand, has shifted from large infrastructure projects under its Belt and Road Initiative to a focus on foreign direct investment in Latin America's critical mineral and EV sectors.[119] It is now focused more on securing long-term supply contracts and investing in the entire value chain of critical minerals, from extraction to refining, and usually does so through the full acquisition of local mines, partnering with local governments, or purchasing significant shares of Western mining companies.[120]

Still, China is increasingly aware that to further gain access to Latin America's array of critical minerals, it might need to further support the development of local supply chains and create value added in the region. It seems Chinese companies are willing to do this. In 2024, for example, BYD announced plans to comprehensively integrate its supply chain in Brazil,[121] signalling increasing Chinese awareness that securing long-term partnerships may require investment in local industries beyond just mineral extraction. BYD is just one of three Chinese automotive companies with plans to make EVs in the country.[122]

While this approach can help some countries diversify their economies, it may also result in further concentration of critical resources under Chinese-controlled firms. This indicates the need for Latin American governments to carefully balance their development goals and Chinese foreign direct investment with maintaining productive relationships with other potential partners, such as the US and the EU.[123] Importers may need to diversify sources, but exporters equally need to diversify offtakers.

Great power competition ultimately serves Latin America: if it has multiple bidding options, it might just be able to get better pricing and outcomes.[124] This causes some Latin American countries to welcome the geopolitical stress and regard it as an opportunity to raise investment for their respective clean energy industries.[125] This is true as long as they can remain nonaligned and not get dragged too much into the deteriorating relations between the US and China. To further attract investments, most countries in the region may consider policies that match their economic targets with those of their desired investors, such as technology sharing through joint ventures and targeted seed funding to universities and companies to move up the value chain in key industries.[126]

Social and environmental demands

While the opportunities are large from an economic standpoint, there is a threat of social discontent around increased mining. While Latin America's mining sector is integral to the global supply chain of critical minerals, it also faces a long history of environmental degradation and social conflict. The 2019 Brumadinho disaster in Brazil, where the collapse of a dam killed 270 people and caused widespread environmental damage, is a stark reminder of the risks associated with irresponsible mining practices.[127] Similarly, 73 per cent of planned and existing critical mineral mines in Latin America are located on or near indigenous or small-farmer land, which has led to rising tensions.[128]

[117] Christopher Hernandez-Roy, Henry Ziemer, Nathaniel Laske, 'De-risking Critical Mineral Supply Chains: The Role of Latin America', CSIS https://www.csis.org/analysis/de-risking-critical-mineral-supply-chains-role-latin-america
[118] World Bank Group, 'GDP (current US$) – Chile', https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=CL
[119] Ilaria Mazzocco, Ryan C. Berg, Rubi Bledsoe, 2024, 'Driving Change : how EVs are reshaping China's economic relationship with Latin America', CSIS, https://www.csis.org/analysis/driving-change-how-evs-are-reshaping-chinas-economic-relationship-latin-america
[120] J.P. Morgan, 2024, 'Global race for critical minerals… a unique opportunity for latin America', https://privatebank.jpmorgan.com/latam/en/insights/markets-and-investing/global-race-for-critical-minerals-a-unique-opportunity-for-latin-america; Ilaria Mazzocco, Ryan C. Berg, Rubi Bledsoe, 2024, 'Driving Change : how EVs are reshaping China's economic relationship with Latin America', CSIS https://www.csis.org/analysis/driving-change-how-evs-are-reshaping-chinas-economic-relationship-latin-america
[121] Christopher Hernandez-Roy, Henry Ziemer, Nathaniel Laske, 'De-risking Critical Mineral Supply Chains: The Role of Latin America', CSIS, https://www.csis.org/analysis/de-risking-critical-mineral-supply-chains-role-latin-america#:~:text=Chinese%20automaker%20BYD-,announced,-plans%20to%20comprehensively
[122] Ilaria Mazzocco, Ryan C. Berg, Rubi Bledsoe, 2024, 'Driving Change : how EVs are reshaping China's economic relationship with Latin America', CSIS, https://www.csis.org/analysis/driving-change-how-evs-are-reshaping-chinas-economic-relationship-latin-america
[123] Ilaria Mazzocco, Ryan C. Berg, Rubi Bledsoe, 2024, 'Driving Change : how EVs are reshaping China's economic relationship with Latin America', CSIS, https://www.csis.org/analysis/driving-change-how-evs-are-reshaping-chinas-economic-relationship-latin-america
[124] Christina Lu, Rocio Fabbro, 2023, 'China's Latin America Gold Rush is All About Clean Energy', Foreign Policy, https://foreignpolicy.com/2023/02/27/china-latin-america-lithium-clean-energy-trade-investment/
[125] Ilaria Mazzocco, Ryan C. Berg, Rubi Bledsoe, 2024, 'Driving Change : how EVs are reshaping China's economic relationship with Latin America', CSIS, https://www.csis.org/analysis/driving-change-how-evs-are-reshaping-chinas-economic-relationship-latin-america
[126] Christopher Hernandez-Roy, Henry Ziemer, Nathaniel Laske, 'De-risking Critical Mineral Supply Chains: The Role of Latin America', CSIS , https://www.csis.org/analysis/de-risking-critical-mineral-supply-chains-role-latin-america
[127] Fangyuan Zhum Wangcheng Zhang, Alexander M. Puzrin, 'The slip surface mechanism of delayed failure of the Brumadinho tailings dam in 2019', Nature, https://doi.org/10.1038/s43247-023-01086-9
[128] Christopher Hernandez-Roy, Henry Ziemer, Nathaniel Laske, 'De-risking Critical Mineral Supply Chains: The Role of Latin America', CSIS, https://www.csis.org/analysis/de-risking-critical-mineral-supply-chains-role-latin-america

As global demand for critical minerals such as lithium and copper grows, Latin American countries are increasingly seeking responsible investors and miners that adhere to higher environmental, social, and governance criteria. Whether or not Latin American mining partners, Chinese or otherwise, can deliver on more responsible mining will matter a great deal to their ability to benefit from the region's mineral wealth.

Major mining companies like BHP and Rio Tinto have already begun incorporating sustainable practices into their operations, and 21 of the 30 largest metals and mining companies by market capitalization have pledged to reduce emissions.[129] Emerging players, such as Sigma Lithium in Brazil, are pioneering zero-waste, low-emission mining techniques, indicating that responsible mining can both be profitable and have a lower environmental footprint.[130]

The pressure for sustainability is coming not only from local communities and governments, but also from the international market. Socially inclined investors and funds have been seeking lower-emission operations in industry and mining and they are willing to pay a premium for it, creating an opportunity for Latin America to attract investment from corporations seeking to meet global sustainability standards.[131]

The onus is not only on Latin American partners, but also on its governments. To attract more responsible investors, further regulatory and permitting reforms are undoubtedly necessary.[132] Governments must ensure that investment in the region benefits local communities, protects the environment, and helps transition the region into a hub for value-added processing, including the development of recycling industries for batteries and other technologies.[133] Latin America's importance in the global energy transition will hinge on the region's ability to attract responsible and sustainable investments, ensuring that mineral extraction is not only profitable but also environmentally sound.[134]

ASSESSING THE VULNERABILITIES AND RISKS OF CHINA'S ROLE IN WIND POWER: A REPORT FROM SWEDEN

Henrik Wachtmeister

China's role in green technology is attracting growing attention in Europe, driven in large part by concerns ranging from unequal competition to potential security risks. Several policy initiatives to address such issues have already been implemented at both EU and national levels, with more likely on the way. So far, discussions and actions have primarily focused on China's dominant role in solar power, batteries, and electric vehicles, but attention is now also shifting to wind power, where China's role is steadily increasing. With a strong domestic wind industry, European policymakers are worried that it might suffer the same fate as its solar industry, where China now almost completely dominates production throughout the value chain.[135] Chinese ownership of large wind farms in Europe has also raised public concerns.[136]

[129] S&P Global, 'ESG Insider Report : Mining &ESG', https://www.spglobal.com/esg/insights/featured/markets-in-motion/mining-and-esg
[130] J.P. Morgan, 2024, 'Global race for critical minerals… a unique opportunity for latin America', https://privatebank.jpmorgan.com/latam/en/insights/markets-and-investing/global-race-for-critical-minerals-a-unique-opportunity-for-latin-america
[131] S&P Global, 'ESG Insider Report : Mining &ESG', https://www.spglobal.com/esg/insights/featured/markets-in-motion/mining-and-esg
[132] Christopher Hernandez-Roy, Henry Ziemer, Nathaniel Laske, 'De-risking Critical Mineral Supply Chains: The Role of Latin America', CSIS, https://www.csis.org/analysis/de-risking-critical-mineral-supply-chains-role-latin-america
[133] Christopher Hernandez-Roy, Henry Ziemer, Nathaniel Laske, 'De-risking Critical Mineral Supply Chains: The Role of Latin America', CSIS, https://www.csis.org/analysis/de-risking-critical-mineral-supply-chains-role-latin-america
[134] J.P. Morgan, 2024, 'Global race for critical minerals… a unique opportunity for latin America', https://privatebank.jpmorgan.com/latam/en/insights/markets-and-investing/global-race-for-critical-minerals-a-unique-opportunity-for-latin-america
[135] "State of the Union Address by President von Der Leyen," Text, European Commission - European Commission, September 13, 2023, https://ec.europa.eu/commission/presscorner/detail/en/speech_23_4426. "Europe Lost to China on Solar — Now It's about to Do the Same with Wind," POLITICO, August 11, 2023, https://www.politico.eu/article/solar-power-china-europ-now-its-about-to-do-the-same-with-wind/.
[136] Björn Conrad and Genia Kostka, "Chinese Investments in Europe's Energy Sector: Risks and Opportunities?," Energy Policy 101 (February 1, 2017): 644–48, https://doi.org/10.1016/j.enpol.2016.12.016; Wiebke Rabe and Olivia Gippner, "Perceptions of China's Outward Foreign Direct Investment in European Critical Infrastructure and Strategic Industries," International Politics 54, no. 4 (July 1, 2017): 468–86, https://doi.org/10.1057/s41311-017-0044-x; Miguel Otero-Iglesias and Manuel Weissenegger, "Motivations, Security Threats and Geopolitical Implications of Chinese Investment in the EU Energy Sector: The Case of CDP Reti," European Journal of International Relations 26, no. 2 (June 1, 2020): 594–620, https://doi.org/10.1177/1354066119871350.

This article summarises and expands on the findings of a recent report examining these issues.[137] The report is a case study of the vulnerabilities and risks associated with China's role in the Swedish wind energy sector. The study covers both Chinese ownership of wind farms and the use of Chinese-made turbines, as well as the reliance of non-Chinese-made turbines on Chinese supply chains, and an assessment of associated risks. It argues that in Sweden, the risks of Chinese ownership of wind farms, which has attracted considerable attention, are likely overblown. Instead, dependence on Chinese supply chains is the most significant short- and long-term vulnerability.

Before discussing results for the case of Sweden in greater depth, this article briefly discusses the framework used for the assessment, in an attempt to make it more useful for application for other cases, and to draw broader policy implications relevant for decision-makers across Europe.

A framework for risk assessment

The report uses a simple risk assessment framework where a distinction is made between *threat*, *vulnerability,* and *risk*, where risk is further operationalised as *specific risk*, *consequence*, *likelihood,* and *risk level*.

A *threat* is defined as an entity that can exploit a vulnerability, and a *vulnerability* is defined as a circumstance or weakness that can be exploited. A vulnerability can be exploited in various ways that pose different kinds of *specific risk*. To assess the magnitude of each specific risk—or the *risk level*—the framework relies on the definition *risk* = *consequence* × *likelihood*, where *consequence* is the (negative) impact of the risk materialising and *likelihood* is the probability of the risk materialising. The final assessed risk level is thus dependent on these two underlying components and can be illustrated in a risk matrix. We use a simple three-step scale of Low, Medium, and High to rate consequence and likelihood and the resulting risk level.

In the Swedish case the threat under consideration is the Chinese state, either directly as an actor with antagonistic intent against Sweden or the EU, or indirectly because of, for example, potential logistical bottlenecks, a potential trade war between the US and China, or an actual war over Taiwan.

The two vulnerabilities assessed are Chinese ownership of wind farms and dependence on Chinese wind energy supply chains. These vulnerabilities can be exploited in different ways and pose several types of specific risks. For Chinese ownership, assessed specific risks include:

- **electricity supply cuts and market manipulation -** the use of ownership of electricity generation to reduce or withhold supply to increase electricity prices in the target market, or to attempt to cause system instability or a blackout;
- **information transfer -** potentially damaging transfers of commercial, technological, and security-related information and know-how through both legal and illegal means;
- **indirect influence -** various types of adaptations by economic and political decision-makers due to relationships with China, such as self-censorship or concessions.

For dependence on Chinese supply chains, assessed specific risks include:

- **export restrictions -** limitations or even complete stops of exports of materials, components, or final products from China or from Chinese companies;
- **IT sabotage -** antagonistic cyber-attacks, which for wind power include malicious code or hacking that can shut down production or even cause physical damage, for example by altering steering mechanisms;
- **indirect influence -** similar to that described above for ownership, but with a different source of influence.

Current and future risks

The likelihood and consequence of a risk materialising depends on several factors. First is the extent of the vulnerability. In the case of ownership, this refers to the share of Chinese ownership, and in the case of supply chains, the level of dependence on Chinese supply chains. A low ownership share or supply chain dependence yields lower consequences, and a high share or dependence yields higher consequences.

[137] Wachtmeister, Henrik. "Chinese Presence in the Swedish Wind Energy Sector: Vulnerabilities and Risks." Stockholm: Swedish National China Centre (NKK), Swedish Institute of International Affairs (UI), 2024. https://kinacentrum.se/en/publications/chinese-presence-in-the-swedish-wind-energy-sector-vulnerabilities-and-risks/.

A second factor altering the likelihood and resulting risk level is the degree of political trust and conflict between China and the analysed entity. For example, if Sweden and China were to develop a trustful relationship comparable to that between Sweden and Norway, the Chinese state would not be perceived as a threat, and subsequent risks would be assessed as low due to the low likelihood of their materialisation.

A third factor, altering the likelihood, is the cost to the threat itself if a specific risk materialises—for example, the economic and reputational cost to China in the case of an export ban.

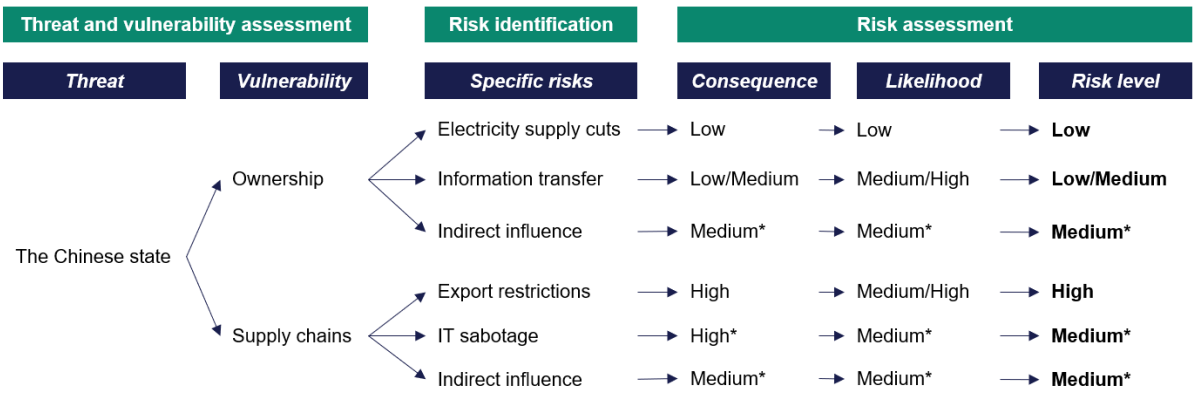
A fourth factor, altering the consequence, is the mitigation or resilience capacity of the targeted entity—for example, in the case of an export ban, the availability of spare parts or substitution options.

Of course, the factors above are not completely independent, and one factor can influence the others. For example, high mitigation capacity likely reduces not only the consequence of a risk materialising, but also the likelihood, since it alters the threat's cost–benefit calculation.

The current setting and relationship with China are complicated. Officially, the EU sees China as a partner for cooperation, an economic competitor, and a systemic rival.[138] The relationship has deteriorated in recent years and some risks have already materialised. Concerns about 'unfair' competition from Chinese firms because of state subsidies and unequal market access have indeed materialised and are seen as a key driver of China's current dominance of supply chains. There is also the longer-term risk of complete dependence on Chinese supply chains. This could open up further specific risks, such as monopolistic practices, and would increase the consequences of (for example) export restrictions significantly. Several of the current specific risks, like information transfer and the (limited) use of export restrictions, can also contribute to the long-term risk of complete market dominance.

The magnitudes of the risks covered in the report were assessed through qualitative analysis and interviews with sector experts. The assessments from low to high should be seen as indicative; they contain a significant degree of uncertainty and primarily reflect the current setting. Figure 1 shows the framework as applied to the Swedish case. The grounds for the rating of some of the key risks are briefly described in the next section.

Figure 1: Outline of the assessment framework as applied in the report



Note: Risk level describes the assessed overall magnitude of the risk, as a function of consequence (the impact of the risk materialising) and likelihood (its probability of materialising); *denotes assessments with a particularly high degree of uncertainty

The Swedish case: ownership and supply chains - survey and risks

This section briefly summarises the quantitative survey of the report and the basis for some of the risk assessments, specifically regarding electricity supply cuts and export restrictions. For the rationale for other specific risk assessments, please refer to the full report.

Chinese companies (all state-owned) control 10.4 per cent of Sweden's installed wind power capacity, which corresponds to 3.4 per cent of total electricity production in 2023 (see Table 1). The majority of these assets are owned by China General Nuclear Power Corporation. The other Chinese ultimate owners are the Silk Road Fund and the State Development and Investment Corporation.

[138] "JOINT COMMUNICATION TO THE EUROPEAN PARLIAMENT, THE EUROPEAN COUNCIL AND THE COUNCIL: EU-China – A Strategic Outlook." Strasbourg: European Commission, 2019. https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52019JC0005. European Parliament, "EU-China Relations: De-Risking or de-Coupling - the Future of the EU Strategy towards China," 2024.

Chinese-made turbines contribute less than 1 per cent of the installed wind capacity, with four mostly small wind farms that came online in 2012, 2014, and 2016, using turbines from Dongfang and Sinovel. According to interviewed experts as well as public reports, these turbines have had some reliability problems in the relatively harsh Swedish climate. This, in contrast to security or other concerns, is likely the reason for the limited use of Chinese turbines. Although no new projects with Chinese turbines are currently under development, newer models have likely improved and are now being offered to European developers at significantly lower prices than those of Western manufacturers. This makes it more likely that the use of Chinese turbines will increase going forward, absent any policy interventions.

The dependence of non-Chinese turbine manufacturers on Chinese supply chains is found to be high, even if is difficult to estimate the precise level from open sources. Globally, China provides 70–80 per cent of many key components, with 80 per cent of the production capacity for gearboxes, 73 per cent for generators, 82 per cent for converters, and 82 per cent for castings. Furthermore, China undertakes around 70 per cent of mining and nearly 100 per cent of the refining of key rare earth elements such as neodymium, used in permanent magnets. According to the Global Wind Energy Council, China generated 64 per cent of the total value across global wind supply chains, from mining to installation, in 2023.[139] However, China is also expected to install 58 per cent of all new turbines. Most of its capacity is thus serving the domestic market (the world's largest) even as it is a net exporter across the value chain. Europe has a strong wind industry, and is self-sufficient in some areas, but is a net importer over the whole value chain.

Table 1: Chinese ownership of Swedish wind farms (names of Chinese-owned companies in bold)

Project name	Location	Capacity (MW)	Production (GWh/year)	Ownership	Acquisition	Seller
Markbyggden Ett	Pitea	644	2,150	**CGN** 75%, **BNR Infrastructure** 25%	2018 & 2020	Green Investment Group & General Electric
Bergvind Lingbo (Valhalla)	Ockelbo	244	751	**CGN** 80%, Hermes 20%	2019	Aquila Capital
Vindpark Tönsen (Valhalla)	Bollnäs	113	349	**CGN** 80%, Hermes 20%	2019	Aquila Capital
Kråktorpet	Sundsvall	163	570	**CGN** 80%, Hermes 20%	2019	Aquila Capital
Lehtirova	Pajala	148	490	**CGN** 80%, Hermes 20%	2019	Aquila Capital
Nylandsbergen	Sundsvall	68	250	**CGN** 80%, Hermes 20%	2019	Aquila Capital
Vindpark Högkölen	Ljusdal	68	250	**CGN** 80%, Hermes 20%	2019	Aquila Capital
Länsterhöjden (Överturingen)	Ånge	84	278	**Red Rock** 50%, CapMan Infra 50%	2020	Green Investment Group
Storflötten (Överturingen)	Ånge	151	500	**Red Rock** 50%, CapMan Infra 50%	2020	Green Investment Group
Total Chinese ownership by share		1,325 (8.2%)	4,890	(14.2% of wind, 3.0% of all production)		
Total Chinese ownership by majority		1,684 (10.4%)	5,588	(16.2% of wind, 3.4% of all production)		

Note: CGN = China General Nuclear Power Corporation
Sources: Svensk Vindenergi[140], Energimyndigheten[141], FOI[142] and other sources listed in Wachtmeister (2024)[143]

[139] GWEC, "Mission Critical: Building the Global Wind Energy Supply Chain for a 1.5°C world", Global Wind Energy Council, December 4, 2023. https://gwec.net/supplychainreport2023/.
[140] Svensk Vindenergi, "Statistik och prognos – Q3 2023, Svensk Vindenergi, 2023. https://svenskvindenergi.org/statistik; Svensk Vindenergi, "Statistik och prognos – Q1 2024", Svensk Vindenergi, 2024. https://svenskvindenergi.org/statistik.
[141] Energimyndigheten. "Vindbrukskollen," 2024. https://vbk.lansstyrelsen.se/.
[142] Oscar Almén, "Kinesiska investeringar i Sverige: en kartläggning", FOI, 2023.
[143] Wachtmeister, Henrik. "Chinese Presence in the Swedish Wind Energy Sector: Vulnerabilities and Risks." Stockholm: Swedish National China Centre (NKK), Swedish Institute of International Affairs (UI), 2024. https://kinacentrum.se/en/publications/chinese-presence-in-the-swedish-wind-energy-sector-vulnerabilities-and-risks/.

In Sweden, even though Chinese ownership of wind farms has attracted considerable attention, the associated risks are likely overblown. This is because the risk of ownership being used for antagonistic supply cuts of electricity under current circumstances is considered low. Since wind power occasionally produces nothing, the system is designed to handle this from a stability perspective. Naturally, reduced production leads to higher prices, all else being equal, but with Chinese entities owning about 10 per cent of the wind capacity, the price effect would be limited. Large production reductions would also be limited to windy days, when other wind production is high and prices often low.

Additionally, all surveyed Chinese producers are bound by Power Purchase Agreements (PPAs). If they do not deliver electricity according to the agreements, incurred debts could lead to bankruptcy and redistribution of assets. The largest Chinese investment in Sweden, China General Nuclear Power Corporation's Markbygden Ett AB, is currently facing economic difficulties with legal proceedings and restructuring following its losses and inability to meet agreed production levels to its PPA counterpart Norsk Hydro. Finally, should stability problems or serious shortages still arise, Svenska Kraftnät, the Swedish TSO (transmission system operator), has the authority to mandate production. In summary, Chinese interests would have much to lose from such actions in terms of reputational and economic costs, while the effect would be limited. The risk is therefor considered low.

However, there are still reasons to consider how much foreign-owned wind power from certain countries Sweden should allow. Additional investment without PPAs removes much of the direct economic costs of withholding supply. Furthermore, if a large enough share were taken offline simultaneously and without warning, it could cause system instability. Naturally, a larger share would also induce a greater price effect. More detailed studies on these risks, ideally conducted together with the TSO, are warranted. A law on the screening of foreign direct investments came into force in December 2023. No Chinese wind acquisitions or new investments have been made since then, and a future potential review will be interesting to follow.

Rather than Chinese ownership, dependence on Chinese supply chains is the most significant short- and long-term vulnerability. Chinese dominance throughout the supply chain could be used to limit the availability of spare parts to existing turbines or components for new turbines and could be employed across a continuum, ranging from subtly skewing competition to complete export bans of components or final products targeted at individual countries or Europe as a whole. The specific risk of export restrictions is therefore rated as high, as a combination of high consequence and medium-to-high likelihood, strongly dependent on future political and economic relations and conflict levels. Export restrictions and supply chain dependence are closely associated with the possibility of the European wind industry being outcompeted on unequal terms. This risks the loss of qualified jobs and industry, and in the long run could put Europe, and Sweden, in a technology dependence that could be used for political influence.

Risk management and policy trilemma

The framework provides some general measures and developments that could reduce risk levels. First, as already discussed, improved relations with the threat country could reduce the likelihood of many risks materialising, as could increased interdependence or deterrence through other means which raise the costs for the threat. However, according to current trajectories, a further deterioration of relations seems more likely than an improvement. What European countries can control, however, is the extent of their vulnerabilities and their capacity to reduce consequences through other means, although this comes with difficult policy trade-offs and associated costs.

Setting a threshold on ownership from 'foreign entities of concern' or banning such investments altogether can reduce risks connected to this vulnerability. Likewise, reducing dependence on Chinese supply chains reduces the risks connected to these. Policies to reduce dependence could range from measures aimed at levelling the playing field to more aggressive trade protectionism and industry-supporting policies. Alternatively, or in addition, countries could attempt to minimise the potential negative impact of risk materialisations. In the case of ownership, this can include clear legal rights for TSOs to intervene in the case of antagonist supply cuts and increase information and IT security. Reducing the impact of export restrictions could involve inventories of spare parts, preparedness to scale up alternative supply chains, or energy technology diversification and substitution options.

Furthermore, it is important to assess the wind power sector risks in relation to overall risks related to China, across all sectors. High dependence, or even complete Chinese dominance, in wind power could perhaps be manageable. However, if wind is only one of several sectors dominated by China, the risks compounds. Particularly if China dominates all green technologies, the risk to Europe's overall energy transition and energy security increases, likely exponentially rather than linearly.

Policies and measures to counter risks inevitably come with costs. The classic energy policy problem is often summarised as the trilemma of affordability, environmental sustainability, and security. There are usually trade-offs between these desirable dimensions. Relying on cheap, large-scale Chinese production can increase both the affordability and speed of the climate transition, but at the cost of security. European policies to support a more secure domestic wind industry are likely to come with a higher economic cost, and perhaps with a slower transition. However, many of the perceived trade-offs are in fact, to a high degree, variable distributions of costs and benefits over time. In many cases, short-term benefits are exchanged for long-term costs. With a long-term perspective, the trilemma could perhaps more easily be squared. A secure, more environmentally sustainable domestic industry could indeed be competitive in the long term.

For example, the complicated logistics required to transport large wind power parts provide a physical competitive advantage for local manufacturers of these parts. Additionally, costly subsidies will be more difficult for China to uphold over the long term. Furthermore, while reliance on Chinese technology could be cost-effective and environmentally beneficial in the short term, it could expose Europe to monopolistic practices in the long term and to dependence on production processes with a dirtier environmental and social footprint, with impacts that accumulate over time.

With the current and potential future risks in mind, and with a long-term view on associated policy trade-offs, measures to ensure fair competition and the survival of the domestic European wind industry could likely be justified. This does not require ceasing trade with China altogether and forgoing the benefits of Chinese companies' technical progress and lower-cost technology. It is likely possible to find a suitable balance for targets and measures to support the domestic wind industry while still allowing and benefiting from Chinese manufacturing and development ability. Further specific discussion on the right balance, including the role of Chinese manufacturing located within Europe, is encouraged going forward. The Carbon Border Adjustment Mechanism is one policy soon to be implemented. Recent industry estimates project significantly increased costs for turbines made in Europe.[144] Concerns also include the mechanism leading to lower imports of raw materials covered in the initial phase and instead increased imports of uncovered processed products in order to circumvent the tax.[145] Consequently, it is necessary to look at the full picture of different industrial policies and their interactions, and to be nimble in addressing distortions and unwanted side effects.

IS CHINA'S SOLAR PV MANUFACTURING DOMINANCE A RISK TO THE UK?

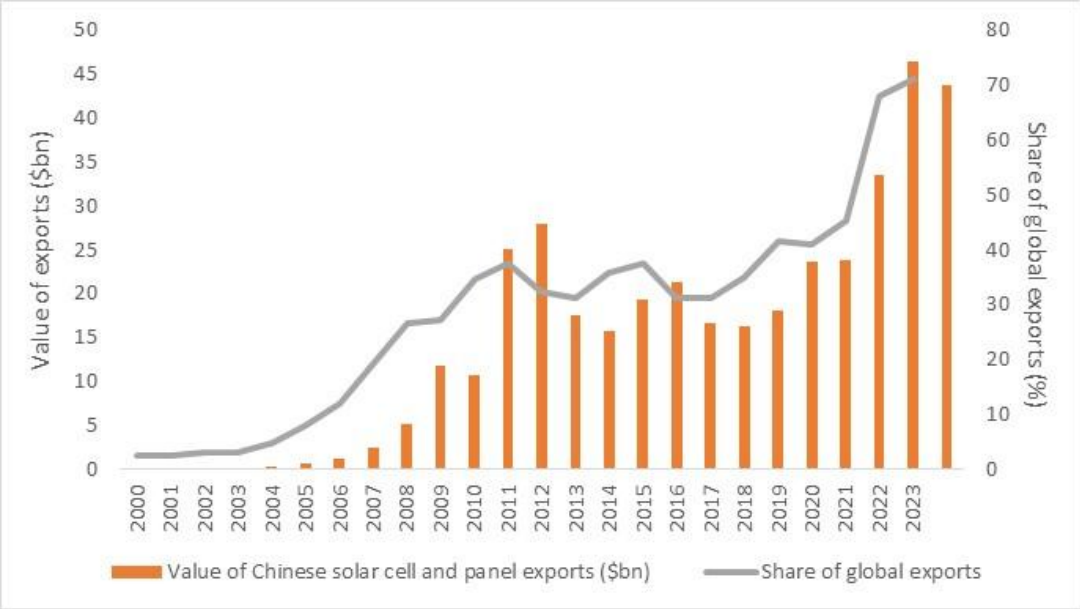
Dan Marks

While the new UK government reviews its relationship with China and related security concerns, it has inherited questions about China's dominance of the solar supply chain from the previous government.[146] Solar supply chain security has prompted a policy response from many of the UK's allies, in particular the United States, but with many of those countries likely to depend much more heavily on solar power than the UK, the extent to which the UK should share this concern has been debated by policymakers in recent years.

It is unquestionable that the global supply of solar PV equipment is dependent on Chinese industry. In 2023, China's exports of solar PV cells, whether assembled into panels or not, was worth $44 billion according to Comtrade, or 71 per cent of all global solar cell exports. That is more than the UK's exports of all passenger vehicles ($37 billion) and significantly more than China's exports of electric vehicles ($34 billion). Chinese exports of solar cells have increased dramatically since 2020 (Figure 1), and solar cells alone accounted for 1 per cent of the value of all China's exports. But what are the risks associated with dependence on China?

[144] Montel. "CBAM May Cost Each Offshore Wind Turbine near EUR 1m – Lobby." MONTEL, October 2, 2024. https://montelnews.com/news/5557280e-68f4-487c-a7b8-6b55ed3ae041/cbam-may-cost-each-offshore-wind-turbine-near-eur-1m-lobby.
[145] Wind Europe. "Setting the Course towards Climate Neutrality: Wind Europe Position on the 'Fit for 55' Package," 2021. https://windeurope.org/wp-content/uploads/files/policy/position-papers/20210930-WindEurope-Setting-the-course-towards-climate-neutrality-fit-for-55-position-paper.pdf.
[146] Abby Wallace, Eleni Courea and Esther Webber, 'The latest Tory worry: China's grip on solar power,' Politico, 17 November 2023, https://www.politico.eu/article/conservative-party-uk-worry-china-solar-power/

Figure 1: Chinese solar cell and panel exports—total value and share of global exports

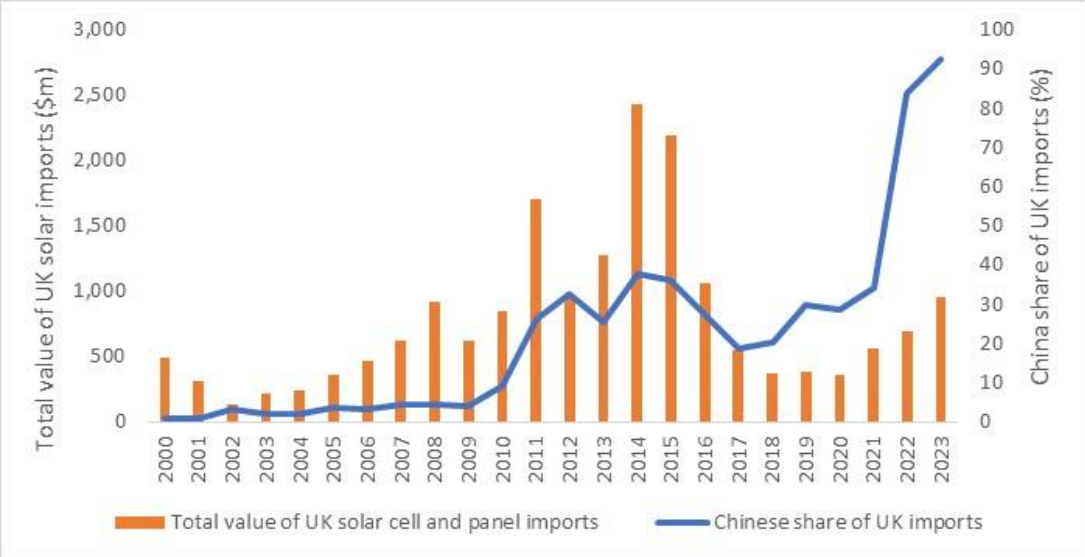


Source: Comtrade (pre-2021 HS code 854140, 2022 onwards HS codes 854142 and 854143), https://comtradeplus.un.org/

The role of solar in the UK's energy transition

In 2022 and 2023, China became a near monopoly supplier of solar cells, panels, and modules to the UK, accounting for 84 per cent and 93 per cent of UK imports respectively in a sector where the UK is entirely dependent on imports. Figure 2 shows that while China suddenly became the dominant exporter to the UK in 2022, this has coincided with a period of middling investment in solar in the UK, compared to the 2011–2015 peak.

Figure 2: UK solar imports—total value and China's share



Source: Comtrade (pre-2021 HS code 854140, 2022 onwards HS codes 854142 and 854143), https://comtradeplus.un.org/

As the UK is not buying massive quantities of solar PV equipment—solar PV imports in 2023 came to just under $1 billion—risks to UK energy security are mostly premised on reliance on the technology in the UK in future.

However, in none of the UK National Energy System Operator's (NESO's) 2024 Future Energy Scenarios does solar PV account for more than 15 per cent of UK electricity generation at any point up to 2050.[147] Solar PV is likely to be one of the easier generation technologies to substitute in the UK: wind will account for the bulk of electricity generation, while gas and storage technologies

[147] National Energy System Operator, 'Future Energy System Data Workbook 2024,' 2024, https://www.neso.energy/fes-data-workbook-2024

will remain central to system balancing and ancillary services. Meanwhile, solar will play a role as a non-dispatchable source of electrons for the grid and improving resilience to businesses and better-off households in small-scale applications, but it will play a secondary role in the resilience and security of the national grid.

On the other hand, even this relatively modest role requires capacity additions of between 19 and 79 GW by 2040 compared to 2023 in the NESO scenarios, and while solar PV may not be critical for energy security, it will play an important role in decarbonisation. Indeed, in scenarios where the UK achieves its 2050 net zero target, solar PV produces 16–74 per cent more electricity in 2050 than was produced using gas-fired combined-cycle power plants in 2023, driven by overall increases in electricity demand. Should energy market volatility persist or worsen during the energy transition, solar PV, particularly in combination with batteries, is likely to be an important tool in allowing households and businesses to mitigate risk.

UK energy security

In this context, an interruption to the supply of solar PV equipment or major price volatility could impact the UK in two ways. It could impact UK energy security and efforts to decarbonise, increasing costs and putting pressure on other supply chains. An interruption would also very directly impact the workers and businesses in the UK's network of solar PV installers, wholesalers, and distributors, and household battery system and software integration providers. A third risk from dependence on imported panels from China is that UK demand contributes to environmental degradation, harm to communities, and negative political drivers around the world.

While the extent of UK energy system technical vulnerability to disruption to solar PV supply chains in China is unclear, the large solar PV capacity required for an optimal energy system in the NESO scenarios suggests that solar price rises or prolonged supply interruptions would have a meaningful economic cost for UK consumers. But the impact on energy security is mitigated by several factors, not least the relatively small share of overall electricity generation expected from solar PV. The UK could deploy wind power, extend the life of gas power plants, or increase imports through planned new interconnectors to compensate for solar supply chain challenges. Solar PV would likely also play only a small role tackling the winter peak, even in scenarios where heating is largely electrified, which is the main energy security challenge in the UK.

There have been attempts to calculate the potential cost to UK electricity consumers of disruptions to solar PV imports from China. These exercises face several challenges. The China Strategic Risks Institute in 2023 attempted to model the cost to the UK of a three- and five-year ban on solar PV exports from China to the UK, finding a significant cost to electricity consumers of £1.5–4.3 billion in total, depending on the timing, duration, and extent of the ban. Whether such a ban would be enforceable in a market where wholesalers play a major role and where global inventories in 2023 rose to 150 GW is doubtful.[148] Furthermore, these models do not typically account for the cost of mitigating the risk, which the US experience shows is significant in terms of both policy support and cost to the consumer.[149]

While risks to operating solar plants from Chinese supply chains appear limited, cyber attacks are possible. Inverters and control devices are potentially vulnerable,[150] which introduces an important element of supply chain risk management to solar PV projects where both the equipment and the operating software are sourced from China. This is particularly the case where the operation of the system is dependent on updates from the software provider for the lifetime of the project. Household systems which are more likely to be internet connected and where supply chain management is weaker may be particularly vulnerable. Military and security sensitive solar applications have an additional range of risks to consider.

The most direct risk to the UK may be to jobs and businesses in the domestic solar industry, which entirely depends on a supply of cost-competitive Chinese panels. As has become clear from the automotive industry, the impact of supply interruptions - or the threats thereof—to industry can be an effective political lever for China. In Europe, the solar industry employed 826,000 people at the end of 2023,[151] suggesting that the industry is a strategic vulnerability for the UK's closest neighbours and allies. The UK industry is much smaller, however, with only around 7,000 people employed in 2020 (less than were employed at the Port Talbot integrated steelworks until recently, and negligible compared with the 758,600 people employed in the automotive sector in 2022[152]), although this could rise to around 60,000 by 2035.[153]

[148] Gaëtan Masson et al, 'Photovoltaic Power Systems Programme: Snapshot 2024,' IEA, 2024, https://iea-pvps.org/snapshot-reports/snapshot-2024/

[149] Hunter Voegele and Ander Ugalde, 'What's in the Inflation Reduction Act?' Ankura, 19 August 2022, https://angle.ankura.com/post/102hv11/whats-in-the-inflation-reduction-act; US produced solar modules cost around 27.5c/W, compared with 11c/W for Chinese panels: Ed Crooks, 'China's solar growth sends module prices plummeting,' Wood Mackenzie, 5 April 2024, https://www.woodmac.com/blogs/energy-pulse/chinas-solar-growth-sends-module-prices-plummeting/

[150] Office of Energy Efficiency and Renewable Energy, 'Solar Cybersecurity Basics,' https://www.energy.gov/eere/solar/solar-cybersecurity-basics

[151] Solar Power Europe, 'EU Solar Jobs Report 2024,' 2024, https://www.solarpowereurope.org/insights/outlooks/eu-solar-jobs-report-2024

[152] Institute of the Motor Industry, 'Driving Forces: Unveiling the Landscape of the K Automotive Labour Market,' 9 August 2023, https://tide.theimi.org.uk/industry-latest/research/uk-automotive-baseline-report-2022-full

[153] Solar Energy UK, 'Join the thriving UK Solar Industry,' https://solarenergyuk.org/join-the-thriving-uk-solar-industry/

Do no harm

Minimising the harm caused by solar PV and related industries and maximising societal benefits is perhaps the most immediate challenge of depending on China for the supply of panels. This is in part because of the challenge of tracing mineral supply chains which converge in Chinese processing facilities. There are also concerns within China. Significant parts of China's solar PV industry are based in Xinjiang and there have been accusations of forced labour in the supply chain.[154] This particularly affects the production of polysilicon, meaning that the supply of panels well beyond China is affected. Indeed, the countries most affected by US legislation banning the use of materials from Xinjiang have been countries in South East Asia, not China.[155]

The carbon intensity of Chinese industry is a challenge, with industry historically attracted to the cheap energy offered by the country's coal regions. Producing solar-grade silicon (silicon with 99.9999999 per cent purity) is a highly energy-intensive process, and more than 75 per cent of global production takes place in China.[156] Around 60 per cent of electricity used in the production of solar panels comes from coal.[157] While the Carbon Border Adjustment Mechanism in the EU may drive Chinese industry towards the country's other low-cost energy source, hydropower, this could exacerbate climate risks to the supply chain from drought, where China is particularly globally vulnerable.[158]

China has the tools to address many of these problems, and some progress has been made.[159] The EU has already begun to leverage the scale of the single market to exert pressure on Chinese companies to improve standards. There is anecdotal evidence that this is having an impact: the fact that Chinese-owned Volvo was the first company in Europe to introduce a battery passport for consumers[160] emphasises both the impact of EU regulation and China's capacity to adapt quickly to it. There are more levers the UK, in concert with Europe, could pull to both pressure Chinese institutions and work with Chinese companies to improve performance.

Environmental, social, and governance challenges associated with the expansion of mining and industry in countries without robust governance frameworks is a potentially stickier challenge. The emergence of a solar PV manufacturing industry in Indonesia, alongside a massive expansion of nickel production, is a case in point.[161] Conventional solar PV uses so-called critical minerals less intensively than other clean energy technologies, but there is likely to be pressure on silver supply chains, where by 2030 demand from solar PV industries could be equivalent to 30 per cent of 2020 global production.[162]

Geopolitics

The small share of projected UK electricity generation and relatively small number of jobs in the industry mean that direct UK risk exposure resulting from dependence on imported Chinese solar panels is limited, at least in the absence of major cyber vulnerabilities to operational plants. But China's overwhelming dominance of the global supply chain should remain a concern for the UK. The International Energy Agency expects that solar could become the world's largest source of electricity by 2035, accounting for 25–35 per cent of total electricity generation, with capacity quadrupling by 2030.[163] China will be at the heart of this transformation.

This has many implications for geopolitics:

- Global decarbonisation depends on China and Chinese companies—nobody else will be able to replicate the country's scale and price competitiveness in this time period.

[154] Anti-slavery, 'Supporting a just transition through sustainable and ethical production of green technologies,' 23 January 2024, https://www.antislavery.org/latest/supporting-a-just-transition-through-sustainable-and-ethical-production-of-green-technologies/
[155] US Customs and Border Protection, 'Uyghur Forced Labor Prevention Act Statistics,' https://www.cbp.gov/newsroom/stats/trade/uyghur-forced-labor-prevention-act-statistics
[156] Jordan Lindsay, 'The Environmental Impact of Manufacturing Solar Photovoltaics (PV),' Minviro, https://www.minviro.com/resources/guides/environmental-impact-of-manufacturing-solar-photovoltaics
[157] Jordan Lindsay, 'The Environmental Impact of Manufacturing Solar Photovoltaics (PV),' Minviro, https://www.minviro.com/resources/guides/environmental-impact-of-manufacturing-solar-photovoltaics
[158] TMP Public, 'Building resilience for critical transition mineral supply chains,' 18 June 2024, https://asktmp.com/wordpress/wp-content/uploads/2024/09/TMP_Building-resilience-for-critical-transition-mineral-supply-chain_JULY24_update.pdf
[159] Cathy Haenlein, Dan Marks and Belinda Schäpe, 'The UK's Role in Advancing Responsible Transition Minerals Supply Chains,' RUSI Occasional Paper, 29 October 2024, https://www.rusi.org/explore-our-research/publications/occasional-papers/uks-role-advancing-responsible-transition-minerals-supply-chains
[160] Lawrence Ulrich, 'Volvo's Battery Passport Traces the Supply Chain,' IEEE Spectrum, 14 June 2024, https://spectrum.ieee.org/volvo-ex90-battery-passport
[161] Nikita Sud, 'Victims of the green energy boom? The Indonesians facing eviction over a China-backed plan to turn their island into a solar panel 'ecocity',' The Conversation, 23 October 2023, https://theconversation.com/victims-of-the-green-energy-boom-the-indonesians-facing-eviction-over-a-china-backed-plan-to-turn-their-island-into-a-solar-panel-ecocity-214755
[162] Jordan Lindsay, 'The Environmental Impact of Manufacturing Solar Photovoltaics (PV),' Minviro, https://www.minviro.com/resources/guides/environmental-impact-of-manufacturing-solar-photovoltaics
[163] IEA, 'World Energy Outlook 2024,' October 2024, https://www.iea.org/reports/world-energy-outlook-2024

- Countries with rapidly expanding electricity demand, particularly where this is driven by current supply shortages in developing countries such as in sub-Saharan Africa, will be vulnerable to disruption from China. Where Chinese supply is complemented by financing, solar could more visibly replace or augment coal and hydropower investment in the country's already formidable array of Belt and Road Initiative tools. Indeed, the need to utilise excess capacity could push moves in this direction.
- Countries involved in the solar PV supply chain such as Thailand and Malaysia will become increasingly dependent on China as an investor, technology provider, and customer. Furthermore, investments in politically aligned countries by major Chinese companies can act as a reward, granting employment and economic benefits as well as improving security of supply. Some commentators have viewed recent Chinese investments in battery plants in Hungary—batteries accounted for 4.7 per cent of the country's total exports in 2022, according to Comtrade—as, in part, a reward for political support.[164]

This is not a straightforward replication of the power of oil and gas producers for a transition age. Solar PV is fundamentally different as infrastructure rather than fuel.[165] But the anticipated importance of solar to global energy supply means that China is positioned as a critical supplier for the first major phase of infrastructure build-out for the energy transition. Beyond notions of dependency, this helps erode some of the basis of Western influence in the world as a supplier of critical technologies and finance and, perhaps, associated systems of regulation, standards, and ultimately wider market governance.

What to do?

As discussed, the UK does not have a solar manufacturing industry to protect or aspirations to build one. The country is expecting to undertake a massive expansion of solar capacity, but at no point is solar PV expected to be a mainstay of electricity generation or a backstop for energy security. The potential impact of solar PV supply interruptions or price volatility on UK jobs is currently limited, although this is likely to increase over time, and there could be marginal impacts on the ability of households and businesses to weather future energy crises. The main domestic risks appear to be to the cost of electricity and to the ability of UK consumers and policymakers to mitigate the environmental and social impact of their solar demand.

At the same time, solar PV is the critical technology for decarbonisation globally. This has four central implications for the UK. Firstly, the UK has an interest in the supply chain being secure and cost-effective to combat climate change. Secondly, while the exposure of the UK to risks from China's dominance of the market may be limited, that is not true for many other countries around the world which are heavily dependent on access to Chinese supply chains for their energy security. Thirdly, countries active in the solar PV supply chain could become increasingly dependent on China. Fourthly, the world's largest energy markets in the US and Europe, as well as emerging countries such as India, have a greater incentive than the UK to invest in dealing with the issue and are already putting in place policy programmes to diversify supply chains and support domestic manufacturing.

Given this starting point, it is fair to assume that the UK will not become a manufacturing hub for solar PV anytime soon and will continue to be almost entirely dependent on imports. China's overwhelming share of global production and exports, combined with current overcapacity, means that 'de-coupling' is not realistic if the UK is to achieve its solar objectives. Diversification of the import supply chain both geographically and in terms of corporate suppliers, then, is likely to be the optimum strategy.

The UK will benefit wherever programmes to support solar manufacturing outside of China are able to help competitors produce at scale. UK government support for innovation and participation by small and medium-sized enterprises, strategies adopted to support diversification in sectors such as telecoms and healthcare, are very unlikely to impact market concentration for solar PV, where manufacturing scale and expertise are decisive. Instead, the starting point for policy should be supporting demand and price levels capable of stimulating further investment in non-Chinese suppliers.

This requires managing the impact of overcapacity in China, in particular the damaging effects of price volatility on manufacturing by non-Chinese companies. The cycle of over-capacity and market flooding, whether intentional or otherwise, typical of many Chinese industries responding to the complex interplay of national and local government policies and domestic and global markets, has played a major role in undermining production elsewhere. The steel industry is one obvious example. Addressing market volatility, dumping, and anti-competitive practices may therefore be foundational in supporting demand for more marginal emerging producers and technologies.

[164] Jordyn Dahl, 'Xi Jinping uses cold hard cash to keep Viktor Orbán close,' Politico, 9 May 2024, https://www.politico.eu/article/viktor-orban-xi-jinping-hungary-china-partnership-electric-vehicles-investments-cooperation/
[165] Walt Patterson, 'Infrastructure needed, not fuel,' April 2005, https://www.waltpatterson.org/infnotfuel.pdf

This starting point could be supplemented by policies which help 'level the playing field' in terms of trading practices and standards,[166] and regulations which incentivise or require diversification and more rigorous supply chain risk management. The UK has a range of existing strategies from other sectors it might draw on to support solar PV import diversification. For example, the UK Telecommunications (Security) Act 2021 requires that telecom operators improve third-party risk management processes and that they are able to operate within the UK without relying on any company, equipment, or data from outside of the country.

Improved data collection would help guide policies; transparency is currently notoriously bad in clean energy supply chains. The 2020 Agriculture Act granted the Secretary of State powers to collect data, albeit only relating to activities in the UK, in order to manage risks and market volatility and to improve transparency and fairness in the supply chain. Similar data collection powers for the seven key clean energy technologies, potentially alongside the equivalent of Defra's food security review, would help the industry move towards the formalisation of data collection and supply chain monitoring.

Despite its near complete lack of domestic solar PV manufacturing - or perhaps because of it, not needing to protect UK businesses—and because of the more limited role solar is expected to play in UK energy security compared to that of other countries, the UK should be less vulnerable to any disruption that might result from the concentration of solar PV industries in China. The country can take advantage of some of the cost and scale benefits from China, but should also play its part to support diversification. The UK's main tool to do this is its domestic market and the way it addresses volatility from China's cycles of over-capacity, supply chain data and transparency, responsible sourcing standards, and regulation of third-party and supply-chain risks.

[166] Cathy Haenlein, Dan Marks and Belinda Schäpe, 'The UK's Role in Advancing Responsible Transition Minerals Supply Chains,' RUSI Occasional Paper, 29 October 2024, https://www.rusi.org/explore-our-research/publications/occasional-papers/uks-role-advancing-responsible-transition-minerals-supply-chains

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